# AGR 101 – FUNDAMENTALS OF AGRONOMY AND AGRICULTURAL HERITAGE (1+1)




**Course Teachers**

**Dr. P. Rajesh**

Assistant Professor (Agronomy)

**Dr. I. Chinnamani**

Assistant Professor (Agronomy)








**Syllabus: AGR 101 Fundamentals of Agronomy and Agricultural Heritage (1+1)**


| Lec. No. | Contents | Page No. |
|---|---|---|
| 1. | Agriculture – Definition – Importance and scope - Branches of agriculture<br>Evolution of man and agriculture | 3 |
| 2. | Indian agriculture - Indian economy – National income – per capita income – Agricultural income in GDP - Women in agriculture and empowerment | 6 |
| 3. | Agriculture heritage – Agriculture in ancient India | 11 |
| 4. | Chronological Agricultural technology development in India –*Kambar Aer Ezhupadhu* | 16 |
| 5. | Development of scientific agriculture - National and International Agricultural<br>Research Institutes. | 21 |
| 6. | Agronomy – definition – meaning and scope. | 26 |
| 7. | Agro-climatic zones of India and Tamil Nadu – Agro ecological zones of India | 29 |
| 8. | Midterm Examination | |
| 9. | Crops and major soils - Classification – Economic and agricultural importance in India and Tamil Nadu | 35 |
| 10. | Factors affecting crop production – climatic – edaphic - biotic- physiographic<br>and socio economic factors | 41 |
| 11. | Tillage – Definition – objectives – types of tillage - modern concepts of tillage – main field preparation | 46 |
| 12. | Seeds - Seed rate - Sowing methods - Germination - Crop stand establishment - Planting geometry | 50 |
| 13. | Weeds – Definition – harmful and beneficial effects of weeds – crop weed competition and management of weeds – IWM | 55 |
| 14. | Role of manures and fertilizers in crop production – agronomic interventions for enhancing FUE - Inter cultivation - Thinning - Gap filling and other intercultural operations<br>MANURES | 61 |
| 15. | Irrigation - Time and methods - Modern techniques of irrigation - Drainage and its importance | 64 |
| 16. | Cropping pattern and cropping system - Intensive cropping- Sustainable agriculture – IFS | 69 |
| 17. | Organic / eco-friendly agriculture - Dry farming - Concepts and principles | 73 |

**Lect 1. Agriculture – Definition – Importance and scope - Branches of agriculture - Evolution of man and agriculture –**

**Agriculture**

The term Agriculture is derived from two Latin words **ager** or **agri** meaning **soil** and **cultura** meaning **cultivation**. Agriculture is an applied science which encompasses all aspects of crop production including horticulture, livestock rearing, fisheries, forestry, etc.

**Agriculture is defined as an art, science and business of producing crops and livestock for economic purposes.**

**As an art** it embraces knowledge of the way to perform the operations of the farm in a skillful manner, but does not necessarily include an understanding of the principles underlying the farm practices.

**As a science:** utilizes all technologies developed on scientific principles such as crop breeding, production techniques, crop protection, economics etc. to maximize the yield and profit. For example, new crops and varieties developed by hybridization, Transgenic crop varieties resistant to pests and diseases, hybrids in each crop, high fertilizer responsive varieties, water management, herbicides to control weeds, use of bio-control agents to combat pest and diseases etc.

**As the business:** As long as agriculture is the way of life of the rural population production is ultimately bound to consumption. But agriculture as a business aims at maximum net return through the management of land labour, water and capital, employing the knowledge of various sciences for production of food, feed, fibre and fuel. In recent years, agriculture is commercialized to run as a business through mechanization.

AGRICULTURE is defined in the Agriculture act (1947), as including ‘horticulture, fruit growing, seed growing, dairy farming and livestock breeding and keeping, the use of land as grazing land, meadow land, osier land, market gardens and nursery grounds, and the use of land for woodlands where that use ancillary to the farming of land for Agricultural purposes”.

## SCOPE AND IMPORTANCE OF AGRICULTURE IN INDIA AND TAMILNADU

- With a 16% contribution to the gross domestic product (GDP), agriculture still provides livelihood support to about two-thirds of country's population.
- The sector provides employment to 58% of country's work force and is the single largest private sector occupation.
- Agriculture accounts for about 15% of the total export earnings and provides raw material to a large number of Industries (textiles, silk, sugar, rice, flour mills, milk products).
- Rural areas are the biggest markets for low-priced and middle-priced consumer goods, including consumer durables and rural domestic savings are an important source of resource mobilization.
- The agriculture sector acts as a wall in maintaining food security and in the process, national security as well.
- The allied sectors like horticulture, animal husbandry, dairy and fisheries, have an important role in improving the overall economic conditions and health and nutrition of the rural masses.
- To maintain the ecological balance, there is need for sustainable and balanced development of agriculture and allied sectors.
- Agriculture's eyes and minds are soothed by dynamic changes from brown (bare soil) to green (growing crop) to golden (mature crop) and bumper harvests.
- Plateauing of agricultural productivity in irrigated areas and in some cases the declining trend warrants attention of scientists.

Agriculture helps to elevate the community consisting of different castes and communities to a better social, cultural, political and economical life. Agriculture maintains a biological equilibrium in nature. Satisfactory agricultural production brings peace, prosperity, harmony, health and wealth to individuals of a nation by driving away distrust, discord and anarchy.

**REVOLUTIONS IN AGRICULTURE**

- Through white revolution, milk production quadrupled from 17 million tonnes at independence to 108.5 million tonnes.
- Through blue revolution, fish production rose from 0.75 million tonnes to nearly 7.6 million tonnes during the last five decades.
- Through yellow revolution oil seed production increased 5 times (from 5 million tonnes to 25 million tonnes) since independence.

- Similarly, the egg production increased from 2 billion at independence to 28 billion, sugarcane production from 57 million tonnes to 282 million tonnes, cotton production from 3 million bales to 32 million bales which shows our sign of progress.
- India is the largest producer of fruits in the world. India is the second largest producer of milk and vegetable.

## BRANCHES OF AGRICULTURE

Seven branches viz.,

1. Agronomy
2. Horticulture
3. Forestry
4. Animal husbandry
5. Fishery science
6. Agricultural Engineering and
7. Home science

**1) Agronomy** – Deals with the production of various crops which includes food crops, fodder crops, fibre crops, sugar, oilseeds, etc. The aim is to have better food production and how to control the diseases.

**2) Horticulture** - Deals with the production of fruits, vegetables, flowers, ornamental plants, spices, condiments and beverages.

**3) Forestry** – Deals with production of large scale cultivation of perennial trees for supplying wood, timber, rubber, etc. and also raw materials for industries.

**4) Animal husbandry** – Deals with agricultural practice of breeding and raising livestock in order to provide food for humans and to provide power (draught) and manure for crops.

**5) Fishery science** – Deals with practice of breeding and rearing fishes including marine and inland fishes, shrimps, prawns etc. in order to provide food, feed and manure.

**6) Agricultural Engineering** – Deals with farm machinery for filed preparation, inter-cultivation, harvesting and post harvest processing including soil and water conservation engineering and bio-energy.

**7) Home Science** – Deals with application and utilization of agricultural produces in a better manner in order to provide nutritional security, including value addition and food preparation.

On integration, all the seven branches, first three is grouped as for crop production group and next two animal management and last two allied agriculture branches.

**Evolution of man and Agriculture**

There are different stages in development of agriculture, which is oriented with human civilization. They are Hunting → Pastoral → Crop culture → Trade (stages of human civilization).

**1. Hunting** – It was the primary source of food in old days. It is the important occupation and it existed for a very long period.

**2. Pastoral** – Human obtained his food through domestication animals, e.g. dogs, horse, cow, buffalo, etc. They lived in the periphery of the forest and they had to feed his domesticated animals. For feeding his animals, he would have migrated from one place to another in search of food. It was not comfortable and they might have enjoyed the benefit of staying in one place near the river bed.

**3. Crop culture** - By living near the river bed, he had enough water for his animals and domesticated crops and started cultivation. Thus he has started to settle in a place.

**4. Trade** – When he started producing more than his requirement the excess was exchanged, this is the basis for trade. When agriculture has flourished, trade developed. This lead to infrastructure development like road, routes, etc.

Agriculture became civilized from crop culture stage. Some important events for different periods that lead to development of scientific agriculture.

| Period | Events |
|---|---|
| Earlier than 10000 BC | Hunting & gathering |
| 7500 BC | Cultivation of crops- Wheat & Barley |
| 3400 BC | Wheel was invented |
| 3000 BC | Bronze used for making tools |
| 2900 BC | Plough was invented, irrigated farming started |
| 2300 BC | Cultivation of chickpea, cotton, mustard |
| 2200 BC | Cultivation of rice |
| 1500 BC | Cultivation of sugarcane |
| 1400 BC | Use of iron |
| 1000 BC | Use of iron plough |

| | |
|---|---|
| 1500 AD | Cultivation of orange, brinjal, pomegranate |
| 1600 AD | Introduction of several crops to India i.e. potato, tapioca, tomato, chillies, pineapple, groundnut, tobacco, rubber, American cotton |

Agriculture is the primary sector. Other sectors are dependent an agriculture. InIndia major allocation has been done in each five year plan to agriculture. In 8th five year plan yearly 23% of the national budget allocation goes to agriculture and allied agrobased cottage industries run on small scales.

More than 70% of the Indian population (70 millions / 1 billion) depends or involved in agriculture and allied activities. Nearly 40% of the net national product is from agricultural sector. Approximately 35% employment is generated from agriculture, out of which 75% is found in rural areas either directly in indirectly.

In India, food grain production increased almost four fold from about 50 million tonnes at independence to more than 204 million tonnes in 1999-2000 through green revolution. Productivity increased from 522 kg/ha to more than 1500 kg/ha. Per capita availability rose from 395 g/day to 510g in the same period.

Despite variation in the performance of individual crops and regions, total food grain production maintained a growth of 2.7% per annum, which kept ahead of population growth at about 2.2% per annum.

Presently the situation is from food scarcity to food security. Through white revolution milk production quadrupled from 17 million tonnes at independence to 69 million tonnes (1997-98). Through blue revolution fish production rose from 0.75 million tonnes to nearly 5.0 million tonnes during the last five decades.

Through yellow revolution oil seed production increased 5 times (from 5 million tonnes to 25 million tonnes) since independence. Similarly, the egg production increased from 2 billion at independence to 28 billion, sugarcane production from 57 million tonnes to 276 million tonnes, cotton production from 3 million bales to 14 million bales which shows our sign of progress.

**Land utilization statistics**

**In India,**

1. Total geographical area : 328.848 million ha.
2. Total reporting area : 304.300 million ha.
3. Area under cultivation : 143.000 million ha.

4. Total cropped area : 179.750 million ha.
5. Area sown more than once : 36.750 million ha.
6. Area not available for cultivation : 161.300 million ha.
7. Area under forest : 66.400 million ha.

**In Tamil Nadu,**

1. Total geographical area : 13.00 million ha.
2. Total reporting area : 13.00 million ha.
3. Net area sown : 5.74 million ha.
4. Total cropped area : 6.91 million ha.
5. Area sown more than once : 1.17 million ha.
6. Area not available for cultivation : 2.35 million ha.
7. Area under forest : 2.02 million ha.
8. Fallow land : 2.18 million ha.
9. Other uncultivated land : 0.71 million ha.

## Lec. 2. Indian agriculture - Indian economy – National income – per capita income – Agricultural income in GDP - Women in agriculture and empowerment

**Indian Agriculture and Economy:**

Indian Agriculture is one of the most significant contributors to the Indian economy. Agriculture is the only means of living for almost 60% of the employed class in India. The agriculture sector of India has occupied almost 43% of India's geographical area. Agriculture is still the only largest contributor to India's GDP (16%) even after a decline in the same in the agriculture share of India. Agriculture also plays a significant role in the growth of socio-economic sector in India.

In the earlier times, India was largely dependent upon food imports, but the successive story of the agriculture sector of Indian economy has made it self-sufficing in grain production. The country also has substantial reserves for the same. India depends heavily on the agriculture sector, especially on the food production unit after the 1960 crisis in food sector. Since then, India has put a lot of effort to be self-sufficient in the food production and this endeavour of India has led to the Green Revolution. The Green Revolution came into existence with the aim to improve the agriculture in India.

The services enhanced by the Green Revolution in the agriculture sector of Indian economy are as follows:

- Acquiring more area for cultivation purposes
- Expanding irrigation facilities
- Use of improved and advanced high-yielding variety of seeds
- Implementing better techniques that emerged from agriculture research
- Water management
- Plan protection activities through prudent use of fertilizers, pesticides.

All these measures taken by the Green Revolution led to an alarming rise in the wheat and rice production of India's agriculture. Considering the quantum leap witnessed by the wheat and rice production unit of India's agriculture, a National Pulse Development Programme that covered almost 13 states was set up in 1986 with the aim to introduce the improved technologies to the farmers. A Technology Mission on Oilseeds was introduced in 1986 right after the success of National Pulse Development Programme to boost the oilseeds sector in Indian economy. Pulses too came under this programme. A new seed policy was planned to provide entree to superior quality seeds and plant material for fruits, vegetables, oilseeds, pulses and flowers.

The Indian government also set up Ministry of Food Processing Industries to stimulate the agriculture sector of Indian economy and make it more lucrative. India's agriculture sector highly depends upon the monsoon season as heavy rainfall during the time leads to a rich harvest. But, the entire year's agriculture cannot possibly depend upon only one season. Taking into account this fact, a second Green Revolution is likely to be formed to overcome such restrictions. An increase in the growth rate and irrigation area, improved water management, improving the soil quality and diversifying into high value outputs, fruits, vegetables, herbs, flowers, medicinal plants and bio-diesel are also on the list of the services to be taken by the Green Revolution to improve the agriculture in India.

**National income**

National Income is important because of the following reasons,

- To see the economic development of the country.
- To assess the developmental objectives.
- To know the contribution of the various sectors to national income.

Internationally some countries are wealthy, some countries are not wealthy and some countries are in-between. Under such circumstances, it would be difficult to evaluate the performance of an economy. Performance of an economy is directly proportionate to the amount of goods and services produced in an economy. Measuring national income is also important to chalk out the future course of the economy. It also broadly indicates people's standard of living. Income can be measured by Gross National Product (GNP), Gross Domestic Product (GDP), Gross National Income (GNI), Net National Product (NNP) and Net National Income (NNI). The Indian economy is the 12th largest in USD exchange rate terms. India is the second fastest growing economy in the world. India's GDP has touched US$1.25 trillion. The crossing of Indian GDP over a trillion dollar mark in 2007 puts India in the elite group of 12 countries with trillion dollar economy. The tremendous growth rate has coincided with better macroeconomic stability. India has made remarkable progress in information technology, high end services and knowledge process services.

**Agricultural income in GDP**

Agriculture sector contributed 32% in 1990-91, 20% during 2005-06 and around 16% now. Though the contribution of agriculture to the GDP income of India, it is great news that today the service sector is contributing more than half of the Indian GDP. It takes India one step closer to the developed economies of the world. Earlier it was agriculture which mainly contributed to the Indian GDP. The Indian government is still looking up to improve the GDP of the country and so several

steps have been taken to boost the economy. Policies of FDI, SEZs and NRI investment have been framed to give a push to the economy and hence the GDP.

**Agriculture per capita income**

The per capita income of the agriculture sector declines to 1/3 of the national per capita income during the recent years. The per capita income of the agriculture population is estimated around Rs. 10,865 in 2010, which is around 32% of the national per capita income at Rs. 33,802/-. Per capita income of the agriculture population was around half (1/2) at Rs. 5,505 of the national per capita income at Rs. 11,433/- during 1980 however, it came down to around 42% at Rs. 6,652/- of the national per capita income at Rs. 16,020/- during 2000.

**Income Distribution in Agriculture sector**

| Year/ Period | Agriculture Share in GDP | Population Dependent on Agriculture | Agriculture Per Capita (in Rs.) |
|---|---|---|---|
| 1980 | 39 | 70 | 4745 (56%) |
| 1990 | 31 | 65 | 5505 (48%) |
| 2000 | 25 | 59 | 6652 (42%) |
| 2010 | 16 | 58 | 10865 (32%) |

**Women in Agriculture and empowerment**

Women in India now participate in all activities such as education, sports, politics, media, art and culture, service sectors, science and technology, etc. Indira Gandhi, who served as Prime Minister of India for an aggregate period of fifteen years is the world's longest serving woman Prime Minister.

The Constitution of India guarantees to all Indian women equality (Article 14), no discrimination by the State [Article 15(1)], equality of opportunity (Article 16) and equal pay for equal work [Article 39(d)]. In addition, it allows special provisions to be made by the State in favour of women and children [Article 15(3)], renounces practices derogatory to the dignity of women [Article 51(a) (e)], and also allows for provisions to be made by the State for securing just and humane conditions of work and for maternity relief. (Article 42).

The feminist activism in India picked up momentum during later 1970s. Since alcoholism is often associated with violence against women in India, many women groups launched anti-liquor campaigns in Andhra Pradesh, Himachal Pradesh, Haryana, Orissa, Madhya Pradesh and other

states. Many Indian Muslim women have questioned the fundamental leaders' interpretation of women's rights under the Shariat law and have criticized the triple talaq system.

In 1990s, grants from foreign donor agencies enabled the formation of new women-oriented NGOs. Self-help groups and NGOs such as Self Employed Women's Association (SEWA) have played a major role in women's rights in India. Many women have emerged as leaders of local movements. For example, Medha Patkar of the Narmada Bachao Andolan.

The Government of India declared 2001 as the Year of Women's Empowerment (*Swashakti*). The National Policy for the Empowerment of Women came was passed in 2001.

In 2006, the case of a Muslim rape victim called Imrana was highlighted in the media. Imrana was raped by her father-in-law. The pronouncement of some Muslim clerics that Imrana should marry her father-in-law led to widespread protests and finally Imrana's father-in-law was given a prison term of 10 years, The verdict was welcomed by many women's groups and the All India Muslim Personal Law Board.

In 2010 March 9, one day after International Women's day, Rajyasabha passed Women's Reservation Bill, ensuring 33% reservation to women in Parliament and state legislative bodies.

. Adjustments have to be made in traditional gender specific performance of tasks. A woman needs to be physically healthy so that she is able to take challenges of equality. But it is sadly lacking in a majority of women especially in the rural areas. They have unequal access to basic health resources.

Most of the women work in agricultural sector either as workers, in household farms or as wageworkers. Yet it is precisely livelihood in agriculture that has tended to become more volatile and insecure in recent years and women cultivators have therefore been negatively affected. The government's policies for alleviating poverty have failed to produce any desirable results, as women do not receive appropriate wages for their labour. There is also significant amount of unpaid or non-marketed labor within the household. The increase in gender disparity in wages in the urban areas is also quite marked as it results from the employment of women in different and lower paying activities. They are exploited at various levels. They should be provided with proper wages and work at par with men so that their status can be elevated in society.

There is no doubt about the fact that development of women has always been the central focus of planning since Independence. Empowerment is a major step in this direction but it has to be seen in a relational context. A clear vision is needed to remove the obstacles to the path of women's emancipation both from the government and women themselves. Efforts should be directed towards all round development of each and every section of Indian women by giving them their due share.

## Lec. 3. Agriculture heritage – Agriculture in ancient India

### Some definitions

- **History** is the continuous record of past events
- **Heritage** is the inherited values carried from one generation to other generation
- **Agricultural heritage** refers to the values and traditional practices adopted in ancient India which are more relevant for present day system.

### Agriculture Heritage in India

Agriculture in India is not of recent origin, but has a long history dating back to Neolithic age of 7500-6500 B.C. It changed the life style of early man from 'nomadic hunter of wild berries and roots' to 'cultivator of land'. Agriculture is benefited from the wisdom and teachings of great saints. The wisdom gained and practices adopted have been passed down through generations. The traditional farmers have developed the nature friendly farming systems and practices such as mixed farming, mixed cropping, crop rotation etc. The great epics of ancient India convey the depth of knowledge possessed by the older generations of the farmers of India. The modern society has lost sight of the importance of the traditional knowledge which had been subjected to a process of refinement through generations of experience. The ecological considerations shown by the traditional farmers in their farming activities are now-a-days is reflected in the resurgence of organic agriculture.

### Ancient literature

The available ancient literature includes the four Vedas, nine Brahnanas, Aranyakas, Sutra literature, Susruta Samhita, Charaka Samhita, Upanishads, the epices Ramayana and Mahabharata, eighteen Puranas, Buddhist and Jain literature, and texts such as Krishi-Parashara, Kautilya's, Artha-sastra, Panini's Ashtadhyahi, Sangam literature of Tamils, Manusmirit, Varahamihira's Brhat Samhita, Amarkosha, Kashyapiya-Krishisukti and Surapala's Vriskshayurveda. This literature was most likely to have been composed between 6000 BC to 1000 AD. The information related to the biodiversity and agriculture (including animal husbandry) are available in these texts.

### Rigveda

Rigveda is the most ancient literary work of India. It believed that Gods were the foremost among agriculturists. According to Amarakosha (a thesaurus of Sanskrit written by the Jain or

Buddhist scholar Amarasimha**)**, Aryans were agriculturists. Manu and Kautilya prescribed agriculture, cattle rearing and commerce as essential subjects, which the king must learn. According to Patanjali (compiler of the Yoga Sūtras) the economy of the country depended on agriculture and cattle-breeding. Plenty of information is available in 'Puranas', which reveals that ancient Indians had intimate knowledge on all agricultural operations. Some of the well known ancient classics of India are namely, Kautilya's'Arthashastra'; Panini's 'Astadhyayi'; Patanjali's 'Mahabhasya'; Varahamihira's 'Brahat Samhita'; Amarsimha's 'Amarkosha' and Encyclopaedic works of Manasollasa. These classics testify the knowledge and wisdom of the people of ancient period. Technical books dealing exclusively with agriculture were Sage Parashara's 'Krishiparashara' in 1000 A.D. Other important texts are Agni Purana and Krishi Sukti attributed to Kashyap (500 A.D.). Ancient Tamil and Kannada works contain lot of useful information on agriculture in ancient India. Agriculture in India made tremendous progress in the rearing of sheep and goats, cows and buffaloes, trees and shrubs, spices and condiments, food and non-food crops, fruits and vegetables and developed nature friendly farming practices. These practices had social and religious undertones and became the way of life for the people. Domestic rites and festivals often synchronised with the four main agricultural operations of ploughing, sowing, reaping and harvesting.

In the Rigveda, there is reference to hundreds and thousands of cows; to horses yoked to chariots; to race courses where chariot races were held; to camels yoked to the chariots; to sheep and goats offered as sacrificial victims, and to the use of wool for clothing. The famous Cow Sukta indicates that the cow had already become the very basis of rural economy. In another Sukta, she is defined as the mother of the Vasus, the Rudras and the Adityas, as also the pivot of Immortality. The Vedic Aryans appear to have large forests at their disposal for securing timber, and plants and herbs for medicinal purposes appear to have been reared by the physicians of the age, as appears in the Atharva Veda. The farmers' vocation was held in high regard, though agriculture solely depended upon the favours of Parjanya, the god of rain. His thunders are described as food-bringing.

**Plant species**

The four Vedas mention more than 75 plant species, Satapatha Bhrahamna mentions over 25 species, and Charkaa Samhita (300 BC) an Aayurvedic (Indian medicine) treatise-mentions more than 320 plants. Susruta (400 BC) records over 750 medicinal plant species. The oldest book, Rigveda (4000 BC) mentions a large number of poisonous and non poisonous aquatic and terrestrial, and domestic and wild creatures and animals. Puranas mention about 500 species of plants. The science of arbori-horticulture had developed well and has been documented in Surapala's

Vrikshayurveda. Forests were very important in ancient times. From the age of Vedas, protection of forests was emphasized for ecological balance. Kautilya in his Artha Sastra (321-296 BC) mentions that superintendent of forests had to collect forest produce through the forest guards. He provides a long list of trees, varieties, of bamboos, creepers, fibrous plants, drugs and poisons, skins of various animals, etc., that come under the purview of this officer.

**Animals**

The preservation of wild animals was encouraged and hunting as a sport was regarded as detrimental to proper development of the character and personality of the ruler, according to Manu (Manusmriti, 2nd Century BC). Specifically, in the Puranas (300-750 AD) the names of Shalihotra on horses and Palakapya on elephants have been found as experts in animal husbandry. For instance, Garudapurana is a text dealing with treatment of animal disorders while the classical work on the treatment of horses is Aswashastra. One chapter in Agnipurana deals with the treatment of livestock and another on treatment of trees.

**Stages of agriculture development - Era of civilization - Importance of Neolithic civilization**

**Stages of agriculture development**

***12000 to 9500 years ago (~10000-7500 BC)***

- Hunters and food-gathers stage existed.
- Stone implements (microliths) were seen throughout the Indian subcontinent.
- Domestication of dog occurred in Iraq.
- Earliest agriculture was by vegetative propagation (e.g.,bananas, sugarcane, yam, sago, palms, and ginger).

***9500 to 7500 years ago (~7500-5500 BC)***

- Wild ancestors of wheat and barley, goat, sheep, pig, and cattle were found.

***7500 to 5000 years ago (~5500-3000 BC)***

- Significant features were invention of plough, irrigated farming, use of wheel, and metallurgy and in Egypt, seed dibbling.

***5000 to 4000 years ago***

- Harappan culture is characterized by cultivation of wheat, barley and cotton; plough agriculture and bullocks for drought. Indus Valley is the home of cotton.
- Wheeled carts were commonly used in the Indus valley.

- Harappans not only grew cotton but also devised methods for ginning / spinning / weaving.

***4000 to 2000 years ago (~2000 BC-1 AD)***

- In North Arcot, bone / stone tools were found.
- In Nevasa (Maharastra), copper and polished stone axes were used. First evidence of the presence of silk was found at this location.
- At Navdatoli on Narmada river (Nemar, Madhya Pradesh), sickles set with stone teeth were used for cutting crop stalks. Crops grown were wheat, linseed, lentil, urd (black gram), mung bean, and khesari.
- In Eastern India, rice, bananas, and sugarcane were cultivated.

***2000-1500 years ago (~ 1-500 AD)***

- Tank irrigation was developed and practiced widely.
- Greek and Romans had trade with South India; pepper, cloth, and sandal wood were imported by Romans.
- Chola King Karikala (190 AD) defeated Cheras and Pandyas, invaded Srilanka, captured 12000 men and used them as slaves to construct an embankment along the Cauvery, 160 km along, to protect land from floods. He has built numerous irrigation tanks and promoted agriculture by clearing forests.

***1500-1000 years ago (~ 500-1000 AD)***

***The Kanauj Empire of Harshavardhana (606-647 AD)***

- Cereals such as wheat, rice and millets, and fruits were extensively grown. A 60-day variety and fragrant varieties of rice are mentioned.
- Ginger, mustard, melons, pumpkin, onion, and garlic are also mentioned.
- Persian wheel was used in Thanesar (Haryana).

***The kingdoms of South India***

- The kingdoms were of the Chalukyas (Badami), Rashtrakutas (Latur), Pallavas (Kanchi), Pandyas, Hoysals (Helebid), and Kakatiyas (Warangal).
- Cholas ushered in a glorious phase in South Indian in the $10^{th}$ century AD.
- New irrigation systems for agriculture were developed- chain tanks in Andhra in the $9^{th}$ century; and 6.4 km Kaveripak bund.
- Cholas maintained links with China, Myanmar, and Campodia.
- The tank supervision committee (Eri-variyam) looked after the maintenance of a village and regulated the water supply.

***1000-700 years ago (~ 1000-1300 AD)***

- Arab conquest of Sind was during 711-712 AD; Md bin Qaism defeated Dahir, the Hindu king of Sind. Arabs were experts in gardening.
- 1290- 1320AD (Reign of Khiljis): Alauddin Khilji destroyed the agricultural prosperity of a major part of India. He believed in keeping the farmers poor.

**Era of civilization**

It is supposed that man was evolved on earth about 1.7 million years ago. This man was evolved from the monkey who started to move by standing erect on his feet. Such man has been called *Homo erectus* (or) Java man (or) Peking man. Later on Java man transformed into Cro-Magnon and Cro-Magnon into modern man. The modern man is zoologically known as *Homo sapiens* (Homo - Continuous, Sapiens - learning habit). A primitive form of *Homo sapiens*, called Neanderthal man (*Homo sapiens neanderthalensis*), was common in Europe and Asia. After the last glacial period (about 10,000 years ago), modern *Homo sapien sapiens*, began to spread all over the globe.

In the beginning such man had been spending his life wildly, but during the period 8700-7700 BC, they started to pet sheep and goat, although the first pet animal was dog, which was used for hunting. The history of agriculture and civilization go hand in hand as the food production made it possible for primitive man to settle down in selected areas leading to formation of society and initiation of civilization. The development of civilization and agriculture had passed through several stages. Archeologist initially classified the stages as stone age, Bronze and Iron age. Subsequently the scholars spilt up the stone age into Paleolithic period (old stone age), Neolithic age (New stone age) and Mesolithic age (Middle stone age).

Each of three ages, saw distinct improvements. The man fashioned and improved tools out of stones, bones, woods etc. to help them in day-to-day life. They started growing food crops and domesticated animals like cow, sheep, goat, dog etc.

***I. The Stone Age culture :*** *(2,50,000 Bc to 3500 BC)*

The stone age is divided into three periods

A. Palaeolithic period (old stone age)
B. Mesolithic period (middle stone age)
C. Neolithic period (new stone age)

***A. Paleolithic age :*** *Hunters and food gatherers (2,50,000 to 10,000 BC)*

This period is characterized by the food gatherers and hunters. The stone age man started making stone tools and crude choppers. The chipped stone tools and chopped pebbles were used for

hinting, cutting and other purposes. He had no knowledge on cultivation and house building.

The Palaeolithic age in India is divided into three phases according to the nature of stone tools used by the people and according the nature of climate

a) Early or lower palaeolithic (2,50,000 to 1,00,000 BC)
b) Middle palaeolithic (1,00,000 to 40,000 BC)
c) Upper palaeolithic (40,000 to 10,000 BC)

***B. Mesolithic period :*** *Hunters and Herders (10,000 to 3,700 BC)*

The transitional period between the end of the Paleolithic and beginning of the Neolithic is called Mesolithic. It began about 10000BC and ended with the rise of agriculture. This period is characterized by tiny stone implements called microliths. The Mesolithic people lived on hunting, fishing and good-gathering. At later stages, they also domesticated animals. The domestication of the dog was the major achievement of the Mesolithic hunter.

***C. Neolithic Age :*** *Food producers (The beginning of Agriculture)*

The Neolithic age began between 9000 to 7500 BC. Neolithic revolution occurred in Western Asia between 9500 and 8500 years ago mainly in the **Fertilie Cresent** (hilly regions embracing Israel, Jordon, Turkey, Iran, Caspian basin and adjoining Iranian plateau). Neolithic revolution brought a major change in the techniques of food production which gave man control over his environment and saved him from the precarious existence of mere hunting and gathering of wild berries and roots. For the first time, he lived in **settled villages** and apart from security from hunger he had leisure time to think and contemplate.

*The main features of Neolithic culture in India*

1. Neolithic culture denotes a stage in economic and technological development in India
2. Use of polished stone axes for cleaning the bushes
3. Hand made pottery for storing food grains
4. Invented textile, weaving and basketry
5. Cultivation of rice, banana sequence and yams in eastern parts of India
6. Cultivation of millets and pulses in south India
7. Discovery of silk

***II. Bronze Age (Chalcolithic culture) (3000-1700 BC):***

The end of the Neolithic period saw the use of metal. The metal to be used first was copper. The term Chalcolithic (stone–copper phase) is applied to communities using stone implements along with copper and bronze. In more advanced communities, the proportion of copper and bronze

implements is higher than that of stones. The chalcolithic revolution began in Mesopotamia in the fourth millennium B.C. from this area it spread to Egypt, and Indus valley.

*The significant features are*

1. Invention of plough
2. Agriculture shifted from hilly area to lower river valley
3. Flood water were stored for irrigation and canals were dug
4. Irrigated farming started in this period
5. Sowing of seed by dibbling with a pointed stick
6. Use of wheel.
7. Salinity problem and water logging were noticed due to canal irrigation.

### *III. The Iron Age*

The Iron phase in India started after the advent of Aryans. Aryan land was called ***Saptasindhava*** i.e. the land of seven rivers (Sutlej, Beas, Ravi, Chenab, Jhelum, Sindhu and Saraswati). Aryans settle along the rivers.

*The salient features of iron age are*

- The harvesting tool used was **sickle**.
- The crops grown were mainly sesame and sugarcane.
- Iron plough, shear, axe, sickles and hoes were used
- The domestic animals were sheep, goats, dogs, mule, tortoise, cattle breeds and elephants.

**Technological advancements in ancient period**

| Technology | Place of origin | Year |
|---|---|---|
| Seed drills | Mesopotamia | 5500 years ago |
| Irrigation systems | Egypt, Mesopotamia | 5000 years ago |
| Iron plough | China | 2600 years ago |
| Mould board plough | China | 2100 years ago |
| Crop rotation, Horse drawn plough | Europe | AD 1000 |
| Water wheels | England | AD 1066 |

**Lec. 4. Chronological Agricultural technology development in India –*Kambar Aer Ezhupadhu***

Agriculture in India is broadly classified in to five different periods before India's independence.

1. Early history (Before 1500 BCE)
2. Vedic period – Post Maha Janapadas period (1500 BCE – 200 CE)
3. Early Common Era – High Middle Ages (200–1200 CE)
4. Late Middle Ages – Early Modern Era (1200–1757 CE)
5. Colonial British Era (1757–1947 CE)

[Note: BCE - short for "Before the Common Era", "Before the Christian Era", or "Before the Current Era". CE - Common Era, Current Era (Christian Era is, however, also abbreviated AD, for *Anno Domini*]

Indian agriculture began by 9000 BCE as a result of early cultivation of plants and domestication of crops and animals. Settled life soon followed with implements and techniques being developed for agriculture. Double monsoons led to two harvests being reaped in one year. Indian products soon reached the world via existing trading networks and foreign crops were introduced to India. Plants and animals; considered essential to their survival by the Indians, came to be worshiped and venerated.

The middle ages saw irrigation channels reach a new level of sophistication in India and Indian crops affecting the economies of other regions of the world under Islamic patronage. Land and water management systems were developed with an aim of providing uniform growth. Despite some stagnation during the later modern era the independent Republic of India was able to develop a comprehensive agricultural program.

**1. Early history (Before 1500 BCE)**

- **9000 BCE:** Wheat and barley were domesticated in the Indian sub Continent. Domestication of horse, sheep and goat soon followed. This period also saw the first domestication of the elephant.
- **8000-6000 BCE:** Barley and wheat cultivation, along with the domestication of cattle, primarily sheep and goat—was visible in Mehrgarh (Balochistan, now in Pakistan). Agro pastoralism in India included threshing, planting crops in rows—either of two or of six—and storing grain in granaries.

- **5000 BCE:** Agricultural communities became widespread in Kashmir.
- **5000-4000 BCE:** Cotton was cultivated. The Indus cotton industry was well developed and some methods used in cotton spinning and fabrication continued to be practiced till the modern Industrialization of India. A variety of tropical fruit such as mango and muskmelon are native to the Indian sub-continent. The Indians also domesticated hemp, which they used for a number of applications including making narcotics, fibre and oil. The farmers of the Indus Valley grew peas, sesame, and dates. Sugarcane was originally from tropical South Asia and Southeast Asia. Different species likely originated in different locations with *S. barberi* originating in India and *S. edule* and *S. officinarum* coming from New Guinea.
- **5440 BCE:** Wild *Oryza* rice appeared in the Belan and Ganges valley regions of northern India. Rice was cultivated in the Indus Valley civilization.
- **4500 BCE:** Irrigation was developed in the Indus Valley Civilization. The size and prosperity of the Indus civilization grew as a result of this innovation, which eventually led to more planned settlements making use of drainage.
- **3000 BCE**: Sophisticated irrigation and water storage systems were developed by the Indus Valley civilization, including artificial reservoirs at Girnar.
- **2600 BCE:** An early canal irrigation system from Circa.
- **2500 BCE:** Archeological evidence of an animal-drawn plough in the Indus Valley civilization
- **2000 BCE:** Agricultural activity included rice cultivation in the Kashmir and Harrappan regions.

**2. Vedic period-Post Maha Janapadas period (1500 BCE–200 CE)**

- Gupta (2004) finds it likely that summer monsoons may have been longer and may have contained moisture in excess than required for normal food production. One effect of this excessive moisture would have been to aid the winter monsoon rainfall required for winter crops.
- In India, both wheat and barley are held to be *Rabi* (winter) crops and—like other parts of the world—would have largely depended on winter monsoons before the irrigation became widespread. The growth of the *Kharif* crops would have probably suffered as a result of excessive moisture.
- Jute was first cultivated in India, where it was used to make ropes and cordage.

- Some animals—thought by the Indians as being vital to their survival—came to be worshiped.
- Trees were also domesticated, worshiped, and venerated—*Pipal* and *Banyan* in particular.
- Others came to be known for their medicinal uses and found mention in the holistic medical system *Ayurveda*.
- **1000–500 BCE:** There are repeated references to iron. Cultivation of a wide range of cereals, vegetables and fruits is described. Meat and milk products were part of the diet; animal husbandry was important. The soil was ploughed several times. Seeds were broadcasted. Fallowing and a certain sequence of cropping were recommended. Cow dung provided the manure. Irrigation was practiced.
- **322–185 BCE:** The Mauryan Empire categorized soils and made meteorological observations for agricultural use. Other Mauryan facilitation included construction and maintenance of dams, and provision of horse-drawn chariots—quicker than traditional bullock carts.
- **300 BCE:** The Greek diplomat Megasthenes, in his book *Indika*— provides a secular eyewitness account of Indian agriculture.

**3. Early Common Era – High middle ages (200–1200 CE)**

- The Tamil people cultivated a wide range of crops such as rice, sugarcane, millets, black pepper, various grains, coconuts, beans, cotton, plantain, tamarind and sandalwood. Jackfruit, coconut, palm, areca and plantain trees were also known.
- Systematic ploughing, manuring, weeding, irrigation and crop protection was practiced for sustained agriculture. Water storage systems were designed during this period.
- Kallanai ($1^{st}$-$2^{nd}$ century CE), a dam built on river Kaveri during this period, is considered the as one of the oldest water-regulation structures in the world still in use.
- Spice trade involving spices native to India—including cinnamon and black pepper—gained momentum as India starts shipping spices to the Mediterranean.
- Roman trade with India followed as detailed by the archaeological record and the *Periplus of the Erythraean Sea*.
- Chinese sericulture attracted Indian sailors during the early centuries of the Common Era.
- **320-550 CE:** Crystallized sugar was discovered by the time of the Guptas and the earliest reference of candied sugar come from India.
- **647 CE:** Chinese documents confirm at least two missions to India, initiated in, for obtaining technology for sugar-refining.

- **875-1279 CE :** Noboru Karashima's research of the agrarian society in South India during the Chola Empire reveals that during the Chola rule land was transferred and collective holding of land by a group of people slowly gave way to individual plots of land, each with their own irrigation system.
- The growth of individual disposition of farming property may have led to a decrease in areas of dry cultivation.
- The Cholas also had bureaucrats which oversaw the distribution of water—-particularly the distribution of water by tank-and-channel networks to the drier areas.

**4. Late middle ages – Early modern era (1200–1757 CE)**

- The construction of water works and aspects of water technology in India is described in Arabic and Persian works. The diffusion of Indian and Persian irrigation technologies gave rise to irrigation systems which bought about economic growth and growth of material culture.
- Agricultural 'zones' were broadly divided into those producing rice, wheat or millets.
- Rice production continued to dominate Gujarat and wheat dominated north and central India.
- The Encyclopædia Britannica details the many crops introduced to India during this period of extensive global discourse.
- **1556-1605 CE:** Land management was particularly strong during the regime of Akbar the Great under whom scholar-bureaucrat Todarmal formulated and implemented elaborated methods for agricultural management on a rational basis.
- Indian crops—such as cotton, sugar, and citric fruits—spread visibly throughout North Africa, Islamic Spain, and the Middle East.
- Though they may have been in cultivation prior to the solidification of Islam in India, their production was further improved as a result of this recent wave, which led to far-reaching economic outcomes for the regions involved.

**5. Colonial British Era (1757–1947 CE)**

- A number of irrigation canals are located on the Sutlej river.
- Few Indian commercial crops—such as Cotton, indigo, opium, and rice—made it to the global market under the British Raj in India.
- The second half of the $19^{th}$ century saw some increase in land under cultivation and agricultural production expanded at an average rate of about 1% per year by the later $19^{th}$ century.

- Due to extensive irrigation by canal networks Punjab, Narmada valley, and Andhra Pradesh became centers of agrarian reforms.
- The British regime in India did supply the irrigation works but rarely on the scale required.
- Community effort and private investment soared as market for irrigation developed.
- Agricultural prices of some commodities rose to about three times between 1870-1920.
- A rich source of the state of Indian agriculture in the early British era is a report prepared by a British engineer, Thomas Barnard, and his Indian guide, Raja Chengalvaraya Mudaliar, around 1774. This report contains data of agricultural production in about 800 villages in the area around Chennai in the years 1762 to 1766. This report is available in Tamil in the form of palm leaf manuscripts at Thanjavur Tamil University, and in English in the Tamil Nadu State Archives.
- 1871: Government of India created Department of Revenue, Agriculture and Commerce which formed as base for Initiation of Agriculture in India.
- 1880: Famine Commission Report was submitted which was base for inception of Agricultural Department.
- 1881: Separate Department of Agriculture at Centre for Famine relief operations
- 1890 : Dr. J.A. Voelcker appointed as a consulting chemist from Royal Agricultural Society (England) - Laid foundation for agricultural research in India.
- 1892 – 1903 - Appointment of Imperial Agricultural Chemist, Imperial Mycologist and Imperial Entomologist – Base for Beginning of inducting the scientist in Agriculture.
- 1901-05: To enhance agricultural education, Establishment of Agricultural Colleges at Pune, Kanpur, Sabour, Nagpur, Coimbatore and Lyallpur (Now in Pakistan).
- 1905: Establishment of Imperial Agricultural Research Institute (IARI) at Pusa (Bihar)
- 1929: Based on Royal Commission on Agriculture's recommendation (1928), Imperial Council of Agricultural Research (ICAR) was establishment to conduct comprehensive research.
- 1931-47: Indian Lac Cess Committee, Indian Central Tobacco Committee, Indian Central Oilseeds Committee were formed to improve research in various crops.

**Republic of India (1947 CE onwards)**

- Special programs were undertaken to improve food and cash crops supply.
- The Grow More Food Campaign (1940s) and the Integrated Production Programme (1950s) focused on food and cash crops supply respectively.

- 1957: All India Coordinated Maize Improvement Project was initiated (First coordinated project) to exploit maize research (Specifically heterosis).
- Five-year plans of India—oriented towards agricultural development—soon followed.
- 1963: Introduction of semi dwarf wheat varieties from CIMMYT, Mexico Formed basis for green revolution.
- 1966: Introduced semi-dwarf rice varieties TN1 & IR 8 from Taiwan and Philippines respectively is formed as base for green revolution.
- Land reclamation, land development, mechanization, electrification, use of chemicals—fertilizers in particular, and development of agriculture oriented 'package approach' of taking a set of actions instead of promoting single aspect soon followed under government supervision.
- The many 'production revolutions' initiated from 1960s onwards included Green Revolution in India, Yellow Revolution (oilseed: 1986-1990), Operation Flood (dairy: 1970-1996), and Blue Revolution (fishing: 1973-2002) etc.
- 1979: National Agricultural Research Project (NARP) was launched to strengthen the research capabilities of SAUs
- Following the economic reforms of 1991, significant growth was registered in the agricultural sector, which was by now benefiting from the earlier reforms and the newer innovations of Agro-processing and Biotechnology.
- 1998: National Agricultural Technology Project (NATP) was initiated to strengthen the research on location specific problems. Contract farming—which requires the farmers to produce crops for a company under contract—and high value agricultural product increased.
- 2006: National Agricultural Innovative Project (NAIP) was launched for End to end approach for solving problems

***Kambar Aer Ezhupadhu***

The great poet kambar in his book Aer Ezhupadhu explained about the importance of agriculture, tillage practices and farmers role in feeding the world. This book was composed in $12^{th}$ century and contains 70 poems. The poet kambar described the practices of rice cultivation by farmers in wetland namely puddling operation, sowing of rice seeds in the nursery, irrigation, manure application, transplanting of rice seedlings in the main field etc., The kings, poets, rich people and others needed food only through agriculture which was supplied by the farmers only. Sadayappa vallal encouraged kambar for composing this work.

**Lec. 5. Development of scientific agriculture - National and International Agricultural Research Institutes.**

## DEVELOPMENT OF SCIENTIFIC AGRICULTURE IN WORLD

Experimentation technique was started (1561 to 1624) by Francis Bacon. He conducted an experiment and found that water is the principle requirement for plant. If the same crop is cultivated for many times fertility is lost.

Jan Baptiste Van Helmont (1572-1644) was actually responsible for conducting a pot experiment. The experiment is called as 'willow tree experiment'. He took a willow tree of weight 5 pounds. He planted in a pot and the pot contained 200 pounds of soil and continuously monitored for five years by only watering the plant. By the end of $5^{th}$ year, the willow tree was weighing 16 pounds. The weight of soil is 198 pounds. He concluded that water is the sole requirement for plants. The conclusion was erroneous.

In the $18^{th}$ century, Arthur Young (1741-1820) published 'Annals of Agriculture'.

In the beginning of $19^{th}$ century, scientist Jean Senebier (1742-1809), a Swiss naturalist, a historian, gave explanation that increase in the weight of plant was due to the consumption of air. Theodar Desaussure gave the principle theme of photosynthesis.

Liebig is a German scientist and considered as the 'Father of agricultural chemistry'. It was his opinion that the growth of plant was proportional to the amount of mineral substances available in the soil. This is called as 'Liebig law of minimum'.

### Chronological events in scientific agriculture

| | |
|---|---|
| Francis Bacon (1561-1624 A.D) | Found the water as nutrient of plants |
| G.R.Glanber (1604-1668 A.D) | Salt peter(KNO3) as nutrient and not water |
| Jethrotull (1674-1741 A.D) | Fine soil particle as plant nutrient |
| Priestly (1730-1799 A.D) | Discovered the oxygen |
| Francis Home (1775 A.D) | Water, air, salts, fire and oil form the plant nutrients |
| Thomas Jefferson (1793 AD) | Developed mould board plough |
| Theodore de-Saussure | Found that plants absorb $CO_2$ from air & release $O_2$; soil supply $N_2$ |

| | |
|---|---|
| Justus van Liebig (1804- 1873) | German chemist developed "law of minimum " |

"A deficiency or absence of the necessary constituent, all others being present, renders the soil barren for crops for which that nutrient is needed" – It is referred as "Barrle concept"

**Advances in Agriculture in 19$^{th}$ Century**

Following Liebig, an agricultural experiment station was started in Rothamsted in England on 1843 (Old Permanent Manurial Experiment – OPME), it dealt with nutrients. Subsequently many developments took place. In U.S. land grant colleges was started in 19$^{th}$ century. Its objective was to meet the expenditure of the college from the land around the colleges. USDA (United States Department of Agriculture) is responsible for the introduction of herbicides 2,4-D and tractor combine for harvesting and threshing. Under Land Grant College, agriculture oriented teaching, research, extension are expanded. Many international research institutes were started for a specific crop.

- 1857–Michigan State University was established to provide agricultural education at college level.
- Gregor Mendal (1866) discovered the laws of hereditary.
- Charles Darwin (1876) published the results of experiments on cross and self fertilization in plants.
- Thomas Malthus (1898) proposed Malthusian Theory – states that humans would run-out of food for everyone inspite of rapid advance in agriculture due to limited land and yield potential of crops (i.e food may not be sufficient in future for the growing population at this current rate of growth in agriculture)
- Blackman (1905) theory of "optima and limiting factors" states that when a process is conditioned as to it's rapidity by a number of separate factors , the rate of the process is limited by the pace of the slowest factor"
- Mitscherlich (1909) proposed the theory of law of diminishing returns that increase in growth with each successive addition of the limiting element is progressively smaller and the response is curvilinear.
- Wilcox (1929) proposed "inverse yield nitrogen law". It states that the growth or the yielding ability of any crop plant is inversely proportional to the mean nitrogen content in the dry matter.
- Macy (1936) proposed a concept of "Critical Percentages of Plant Nutrients" He suggested a relationship between the sufficiency of nutrients and plant response in terms of yield and nutrient

concentration of plant tissues". Macy proposed critical percentages for each nutrients in each kind of plant.

## DEVELOPMENT OF SCIENTIFIC AGRICULTURE IN INDIA

Scientific agriculture got momentum in the 19$^{th}$ century itself. Indian Land Tax was levied in the middle of 19$^{th}$ century. In 1877, 1878, 1889, 1892, 1897 and 1900, the population was decreased due to continuous famines. Only due to these famines, the British regime started many development programmes. Lord Dalhousie (1848-1856) period the 'Upper Bari Doab Canal' in Punjab was constructed. Improvement of agriculture started only in his period. In Lord Curzon's (1898-1905) period, the 'Great Canal system of Western Punjab' was constructed. During his period Imperial Agricultural Research Institute was started in Pusa in Bihar. His period is called as 'Golden period of agriculture'. During his regime, Department of Agriculture and Agricultural colleges for provinces were started at Coimbatore in 1906.

Due to earthquake at IARI in Pusa, Bihar it was shifted to New Delhi. In 1926, Royal Commission on Agriculture was setup and was responsible for giving recommendation to dug canals, lay roads, etc. Based upon the recommendation of Royal Commission, ICAR (Imperial Council of Agricultural Research) was started in 1929 with the objective to conduct agriculture research. State Agricultural Universities (SAU) were started after 1960s). ICAR had also started research institutes of its own in different centres in India for various crops.

ICAR is the sole body, which controls all the Agricultural Research Institutes in India. It paved way for green revolution in India. After 1947, ICAR totally adapted to Land Grant Colleges. In 1962, a Land Grant College was started in Pantnagar (UP). It is the first university with 16,000 acres.

There are 45 state agricultural universities with research institutes on its own. High yielding wheat varieties like Kalyansona, Sonalika, Lerma roja and Sonara-64 were introduced. Green revolution took in wheat first, next in rice after the invention of Indo-*Japanica* variety. Today, agricultural research is multi-dimensional. In includes tissue culture, biotechnology besides breeding, crop production and crop protection.

### Milestones

- 1880 - Department of Agriculture was established
- 1905 - Imperial Agricultural Research Institute (IARI) was started at Pusa, Bihar

- 1912 - Sugarcane Breeding Institute was established in Coimbatore
- 1929 - Imperial Council of Agricultural Research at New Delhi (then ICAR) after independence becomes ICAR
- 1936 - Due to earth quake in Bihar, IARI was shifted to New Delhi and the place was called with original name Pusa
- 1962- First Agricultural University was started at Pantnagar
- 1965-67 - Green revolution in India due to introduction of HYV –Wheat, rice, use of fertilizers, construction of Dams and use of pesticides

**In Tamil Nadu**

- 1876 - Madras Agricultural College was established at Saidapet
- 1906 - Agricultural College & Research Institute was established at Coimbatore
- 1971 - Tamil Nadu Agricultural University was started

For institutes visit http://www.icar.org.in/node/325

For state agricultural universities visit http://www.icar.org.in/en/universities.htm

**Institutions - 45**

1. Central Rice Research Institute, Cuttack
2. Vivekananda Parvatiya Krishi Anusandhan Sansthan, Almora
3 Indian Institute of Pulses Research, Kanpur
4. Central Tobacco Research Institute, Rajahmundry
5. Indian Institute of Sugarcane Research, Lucknow
6. Sugarcane Breeding Institute, Coimbatore
7. Central Institute of Cotton Research, Nagpur
8. Central Research Institute for Jute and Allied Fibres, Barrackpore
9. Indian Grassland and Fodder Research Institute, Jhansi
10. Indian Institute of Horticultural Research, Bangalore
11. Central Institute of Sub Tropical Horticulture, Lucknow
12. Central Institute of Temperate Horticulture, Srinagar
13. Central Institute of Arid Horticulture, Bikaner
14. Indian Institute of Vegetable Research, Varanasi
15. Central Potato Research Institute, Shimla
16. Central Tuber Crops Research Institute, Trivandrum
17. Central Plantation Crops Research Institute, Kasargod

18. Central Agricultural Research Institute, Port Blair
19. Indian Institute of Spices Research, Calicut
20. Central Soil and Water Conservation Research & Training Institute, Dehradun
21. Indian Institute of Soil Sciences, Bhopal
22. Central Soil Salinity Research Institute, Karnal
23. ICAR Research Complex for Eastern Region including Centre of Makhana, Patna
24. Central Research Institute of Dryland Agriculture, Hyderabad
25. Central Arid Zone Research Institute, Jodhpur
26. ICAR Research Complex Goa
27. ICAR Research Complex for NEH Region, Barapani
28. National Institue of Abiotic Stress Management, Malegaon, Maharashtra
29. Central Institute of Agricultural Engineering, Bhopal
30. Central Institute on Post harvest Engineering and Technology, Ludhiana
31. Indian Institute of Natural Resins and Gums, Ranchi
32. Central Institute of Research on Cotton Technology, Mumbai
33. National Institute of Research on Jute & Allied Fibre Technology, Kolkata
34. Indian Agricultural Statistical Research Institute, New Delhi
35. Central Sheep and Wool Research Institute, Avikanagar, Rajasthan
36. Central Institute for Research on Goats, Makhdoom
37. Central Institute for Research on Buffaloes, Hissar
38. National Institute of Animal Nutrition and Physiology, Bangalore
39. Central Avian Research Institute, Izatnagar
40. Central Marine Fisheries Research Institute, Kochi
41. Central Institute Brackishwater Aquaculture, Chennai
42. Central Inland Fisheries Research Institute, Barrackpore
43. Central Institute of Fisheries Technology, Cochin
44. Central Institute of Freshwater Aquaculture, Bhubneshwar
45. National Academy of Agricultural Research & Management, Hyderabad

**National Research Centres - 17**

1. National Research Centre on Plant Biotechnology, New Delhi
2. National Centre for Integrated Pest Management, New Delhi
3. National Research Centre for Litchi, Muzaffarpur
4. National Research Centre for Citrus, Nagpur

5. National Research Centre for Grapes, Pune
6. National Research Centre for Banana, Trichi
7. National Research Centre Seed Spices, Ajmer
8. National Research Centre for Pomegranate, Solapur
9. National Research Centre on Orchids, Pakyong, Sikkim
10. National Research Centre Agroforestry, Jhansi
11. National Research Centre on Camel, Bikaner
12. National Research Centre on Equines, Hisar
13. National Research Centre on Meat, Hyderabad
14. National Research Centre on Pig, Guwahati
15. National Research Centre on Yak, West Kemang
16. National Research Centre on Mithun, Medziphema, Nagaland
17. National Centre for Agril. Economics & Policy Research, New Delhi

**Important International Institutions on Agricultural Research**

AVRDC- Asian Vegetable Research and Development Centre, Taiwan

CIAT – Centro International de Agricultura Tropical , Cali, Colombia

CIP – Centro International da la Papa ( International potato research institute (Lima, Peru, South America)

CIMMYT – Centro International de Mejoramiento de Maizy Trigo.(International Centre for maize and Wheat development (Londress, Mexico)

IITA –International Institute for Tropical Agriculture, Ibadon in Nigeria, Africa)

ICARDA – International Center for Agricultural Research in the Dry Areas (Aleppo, Syria)

ICRISAT – International Crops Research Institute for the Semi Arid Tropics (Pattancheru in Hyderabad, India)

IIMI- International Irrigation Management Institute, Colombo, SRILANKA

IRRI – International Rice Research Institute (Los Banos, Philippines)

ISNAR- International Service In National Agricultural Research The Hague, Netherlands

WARDA - West African Rice Development Association Ivory coast, Africa.

IBPGR - International Board for Plant Genetic Resources, Rome, Italy

CGIAR – Consultative Group on International Agricultural Research, Washington D.C

FAO – Food and Agricultural Organization, Rome

WMO- World Meteorological Organization, Vienna.

**Lect. 6. Agronomy – definition – meaning and scope.**

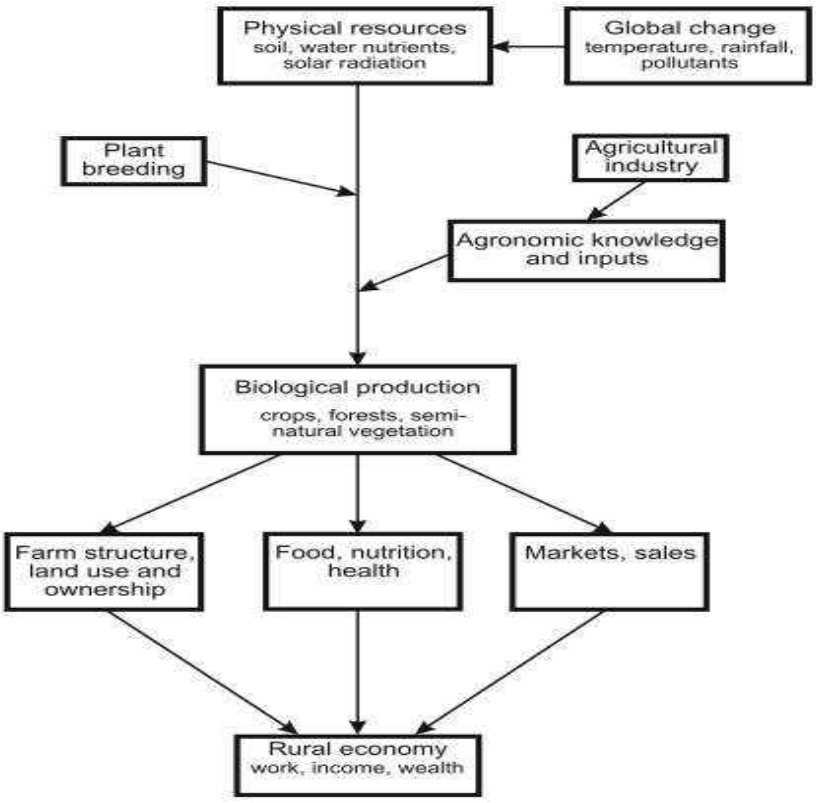
Agronomy is derived from a Greek word '**agros'** meaning 'field' and '**nomos'** meaning 'management'. Principles of agronomy deal with scientific facts in relations to environment in which crop are produced.

**Definition of Agronomy**

1. It is defined as an agricultural science deals with principles and practices of crop production and field management.
2. Agronomy is branch of agricultural science, which deals with principles, & practices of soil, water & crop management.
3. It is branch of agricultural science that deals with methods which provide favorable environment to the crop for higher productively,

**Boundaries and scale**

Crop management, and its scientific study agronomy, are part of a system that comprises the physical elements of the climate, soil and land, the biological constituents of the vegetation and soil, the economic opportunities and constraints of markets, sales and profit, and the social circumstances and preferences of those who work the land.




Flow diagram of physical, biological, economic and social dimensions of agronomy

**Scope of Agronomy**

Agronomy is a dynamic discipline with the advancement of knowledge and better understanding of planet, environment and agriculture. Agronomy science becomes imperative in Agriculture in the following areas.

- Identification of proper season for cultivation of wide range of crops is needed which could be made possible only by Agronomy science.
- Proper methods of cultivation are needed to reduce the cost of cultivation and maximize the yield and economic returns.
- Availability and application of chemical fertilizers has necessitated the generation of knowledge to reduce the ill-effects due to excess application and yield losses due to the unscientific manner of application.
- Availability of herbicides for control of weeds has led to development for a vast knowledge about selectivity, time & method of its application.
- Water management practices play grater role in present day crisis of water demand and Agronomy science answer to the questions ‘how much to apply?’ and ‘when to apply?’.
- Intensive cropping is the need of the day and proper time and space intensification not only increase the production but also reduces the environmental hazards.
- New technology to overcome the effect of moisture stress under dry land condition is explored by Agronomy and future agriculture is depends on dry land agriculture.
- Packages of practices to explore full potential of new varieties of crops are the most important aspects in crop production which could be made possible only by Agronomy science.
- Keeping farm implements in good shape and utilizing efficient manner to nullify the present day labour crisis is further broadening the scope of agronomy.
- Maintaining the ecological balance through efficient management of crops, livestock and their feedings in a rational manner is possible only by knowing agronomic principles.
- Care and disposal of farm and animal products like milk and eggs and proper maintenance of accounts of all transactions concerning farm business is governing principles of agronomy.

**Relation of agronomy to other sciences**

Agronomy is a main branch of Agriculture. It is synthesis of several disciplines like soil science, Agricultural chemistry, crop physiology, plant ecology, biochemistry and economics.

- The Soil Science helps the agronomist to thoroughly understand the soil physical, chemical and biological properties to effect modification of the soil environment.
- The Agricultural Chemistry help the agronomist to understand the chemical composition and changes involved in the production, protection, and use of crops and livestock.
- The crop physiology helps to understand the basic life process of crops to understand functioning of each parts of plant to determine their input requirement like nutrients etc.
- The plant ecology helps us to understand the associated environment in which the crops grown like the influence of weather (Temperature, Rainfall etc).
- The biochemistry shows the way in which biochemical process takes place in crops which helps to understand critical requirements to favourably activate this process.
- The economics paves the way for profit and loss analysis in farming.

**Role of Agronomist**

Agronomist is a scientist who is dealing with the study of problems of crop production and adopting/recommending practices of better field crop production and soil management to get high yield and income.

- Agronomist aims at obtaining maximum production at minimum cost by exploiting the knowledge of the basic and applied sciences for higher crop production.
- In a broader sense, agronomist is concerned with production of food and fibre to meet the needs of growing population.
- He develops efficient and economic field preparation method for sowing crops in different season. (Flat bed, Ridges and furrows)
- He is also involved to selection of suitable crop and varieties to suit or to match varied seasons and soils. Eg. Red soil - groundnut, Black soil - cotton, Sandy soil – tuberous crops, Saline soil – Finger millet (*Ragi*). In *Kharif* if water is sufficient go for rice and water is not sufficient go for maize, sorghum.
- Evolves efficient method of cultivation (whether broadcasting, nursery and transplantation or dibbling, etc.) provides better crop establishment and maintain required population
- He has to identify various types of nutrients required by crops including time and method of application (e.g. for long duration rice (150-60-60 kg NPK), short duration: 120:50:50 kg NPK/ha Application P&K basal and N in three splits)
- Agronomist must select a better weed management practice. Either through mechanical or physical (by human work) or chemical (herbicides or weedicides, e.g. 2-4-D) or cultural (by

having wide space it may increase weed growth by using inter space crops). Weeds are controlled by integrated weed management method also

- Selection of proper irrigation method, irrigation scheduling i.e. irrigation timing and quantity based on the crops to be irrigated, whether to irrigate continuously or stop in between and how much water to be supplied are computed by agronomy science so as to achieve maximum water use efficiency.
- Crop planning (i.e.) suitable crop sequence are developed by agronomist (i.e.) what type of crop, cropping pattern, cropping sequence, etc. (Rice - Rice - Pulse)
- Agronomists are also develops the method of harvesting, time for harvesting, etc. (Appropriate time of harvest essential to prevent yield loss)
- Agronomist is responsible for every decision made in the farm management. (What type of crop to be produced? How much area to be allotted for each crop? How and when to market? How and When to take other management activities?) All the decisions should be taken at appropriate time to efficiently use resources available)

## Lect. 7. Agro-climatic zones of India and Tamil Nadu – Agro ecological zones of India

### Agro-climatic zones

An agro-climatic zone is a land unit uniform in respect of climate and length of growing period (LGP) which is climatically suitable for a certain range of crops and cultivars (FAO, 1983).

### Classification by Planning Commission

Planning Commission of India (1989) made an attempt to delineate the country into different agro climatic regions based on homogeneity in rainfall, temperature, topography, cropping and farming systems and water resources. India is divided into 15 agro-climatic regions.

***1. Western Himalayan zone***

This zone consists of three distinct sub-zones of Jammu and Kashmir, Himachal Pradesh and Uttar Pradesh hills. The region consists of skeletal soils of cold region, podsolic mountain meadow soils and hilly brown soils. Lands of the region have steep slopes in undulating terrain. Soils are generally silty loams and these are prone to erosion hazards.

***2. Eastern Himalayan zone***

Sikkim and Darjeeling hills, Arunachal Pradesh, Meghalaya, Nagaland, Manipur, Tripura, Mizoram, Assam and Jalpaiguri and Coochibihar districts of West Bengal fall under this region, with high rainfall and high forest cover. Shifting cultivation is practiced in nearly one-third of the cultivated area and this has caused denudation and degradation of soils with the resultant heavy runoff, massive soil erosion and floods in lower reaches and basins.

***3. Lower Gangetic Plains zone***

This zone consists of West Bengal-lower Gangetic plain region. The soils are mostly alluvial and are prone to floods.

***4. Middle Gangetic Plains zone***

This zone consists of 12 districts of eastern Uttar Pradesh and 27 districts of Bihar plains. This zone has a geographical area of 16 million hectares and rainfall is high. About 39% of gross cropped area is irrigated and the cropping intensity is 142%.

***5. Upper Gangetic Plains zone***

This zone consists of 32 districts of Uttar Pradesh. Irrigation is through canals and tube wells. A good potential for exploitation of ground water exists.

***6. Trans-Gangetic Plains zone***

This zone consists of Punjab, Haryana, Union territories of Delhi and Chandigarh and Sriganganagar district of Rajasthan. The major characteristics of this area are: highest net sown area, highest irrigated area, high cropping intensity and high groundwater utilization.

***7. Eastern Plateau and Hills zone***

This zone consists of eastern part of Madhya Pradesh, southern part of West Bengal and most of inland Orissa. The soils are shallow and medium in depth and the topography is undulating with a slope of 1-10%. Irrigation is through tanks and tube wells.

***8. Central Plateau and Hills zone***

This zone comprises of 46 district of Madhya Pradesh, part of Uttar Pradesh and Rajasthan. The topography is highly variable nearly 1/3$^{rd}$ of the land is not available for cultivation. Irrigation and cropping intensity are low. 75% of the area is rainfed grown with low value cereal crops. There is an intensive need for alternate high value crops including horticultural crops.

***9. Western Plateau and Hills zone***

This zone comprises the major part of Maharastra, parts of Madhya Pradesh and one district of Rajasthan. The average rainfall of the zone is 904 mm. The net sown area is 65% and forests occupy 11%. The irrigated area is only 12.4% with canals being the main source.

***10. Southern Plateau and Hills zone***

This zone comprises 35 districts of Andhra Pradesh, Karnataka and Tamil Nadu which are typically semi-arid zones. Dryland farming is adopted in 81% of the area and the cropping intensity is 111 percent.

***11. East Coast Plains and Hills zone***

This zone comprises of east coast of Tamil Nadu, Andhra Pradesh and Orissa. Soils are mainly alluvial and coastal sands. Irrigation is through canals and tanks.

***12. West Coast Plains and Ghats zone***

This zone comprises west coast of Tamil Nadu, Kerala, Karnataka, Maharastra and Goa with a variety of crop patterns, rainfall and soil types.

***13. Gujarat Plains and Hills zone***

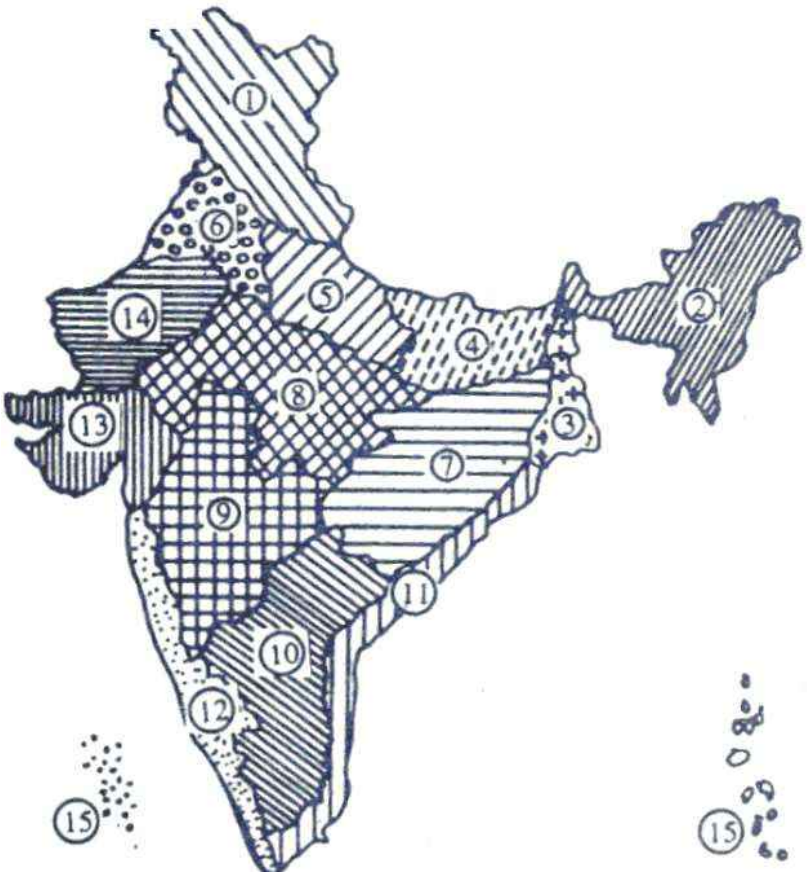
This zone consists of 19 districts of Gujarat. This zone is arid with low rainfall in most parts and only 32.5% of the area is irrigated largely through wells and tube wells.

***14. Western Dry zone***

This zone comprises nine districts of Rajasthan and is characterized by hot sandy desert, erratic rainfall, high evaporation, scanty vegetation. The ground water is deep and often brackish. Famine and drought are common features of the region.

***15. Islands zone***

This zone covers the island territories of Andaman and Nicobar and Lakshadeep which are typically equatorial with rainfall of 3000 mm spread over eight to nine months. It is largely a forest zone with undulated lands.




| | | |
|---|---|---|
| 1 | Western Himalayan Region | J&K, HP, UP, Utranchal |
| 2 | Eastern Himalayan Region | Assam Sikkim, West Bengal & North-Eastern states |
| 3 | Lower Gangetic Plains Region | West Bengal |
| 4 | Middle Gangetic Plains Region | UP, Bihar |
| 5 | Upper Gangetic Plains Region | UP |
| 6 | Trans-Gangetic Plains Region | Punjab, Haryana, Delhi & Rajasthan |
| 7 | Eastern Plateau and Hills Region | Maharastra, UP, Orissa & West Bengal |
| 8 | Central Plateau and Hills Region | MP, Rajasthan, UP |
| 9 | Western Plateau and Hills Region | Maharastra, MP & Rajasthan |
| 10 | Southern Plateau and Hills Region | AP, Karnataka, Tamil Nadu |
| 11 | East Coast Plains and Hills Region | Orissa, AP, TN,& Pondichery |

| | | |
|---|---|---|
| 12 | West Coast Plains and Ghat Region | TN, Kerala, Goa, Karnataka, Maharastra |
| 13 | Gujarat Plains and Hills Region | Gujarat |
| 14 | Western Dry Region | Rajasthan |
| 15 | The Islands Region | Andman & Nicaobar, Lakshya Deep |

**Classification by ICAR**

The State Agricultural Universities were advised to divide each state into sub-zones, under the National Agricultural Research Project (NARP) under ICAR. Based on the rainfall pattern, cropping pattern and administrative units, 127 agro-climatic zones are classified. The zones of each state are given below.

| State | No. of zones | State | No. of zones |
|---|---|---|---|
| Andhra Pradesh | 7 | Madhya Pradesh | 12 |
| Assam | 6 | Rajasthan | 9 |
| Bihar | 6 | Maharashtra | 9 |
| Gujarat | 8 | North Eastern Hill region | 6 |
| Haryana | 2 | Orissa | 9 |
| Himachal Pradesh | 4 | Punjab | 5 |
| Jammu and Kashmir | 4 | Tamil Nadu | 7 |
| Karnataka | 10 | Uttar Pradesh | 10 |
| Kerala | 8 | West Bengal | 6 |

The state of Tamil Nadu has been classified into seven distinct agro-climatic zones listed below.

1. North Eastern zone
2. North Western zone
3. Western zone
4. Cauvery Delta zone
5. Southern zone
6. High Rainfall zone
7. Hilly zone

***1. North Eastern zone***

This zone covers the districts of Thiruvallur, Vellore, Kanchipuram, Thiruvannamalai, Viluppuram, Cuddalore (excluding Chidambaram and Kattumannarkoil taluks), some parts of

Perambalur including Ariyalur taluks and also Chennai. The mean annual rainfall of this region is 1054 mm received in 53 rainy days and is benefited by both the monsoons. The mean monthly maximum temperature ranges between 28.2 to 38.9 °C and the minimum ranges from 19.5 to 24.8°C.

***2. North Western zone***

This zone comprises of Dharmapuri and Krishnagiri district (excluding hilly areas), Salem, Namakkal district (excluding Tiruchengode taluk) and Perambalur taluk of Perambulur district. The climate prevailing in this region is dry and sub humid. This region has been identified as moderately drought prone area. The elevation varies from 330 to 1070 m above mean sea level. The mean annual rainfall of this region is appreciably lower than in North Eastern zone and is 825 mm received in 47 rainy days. The region is benefited by both south-west and north-east monsoon rains but unlike the NEZ, the former contributed more to the total rainfall. The mean monthly maximum temperature ranges between 30 to 37°C with minimum temperature ranging between 19 to 25.5°C. The annual PET of this region is 1727 mm compared to the annual precipitation of 825 mm.

***3. Western zone***

This zone comprises of Erode, Coimbatore, Dindigal, Theni districts, Tiruchengode taluk of Namakkal district, Karur taluk of Karur district and some western part of Madurai district. The mean annual rainfall is 718 mm in 45 rainy days. The monthly mean maximum temperature is 35°C in April and 30°C in January and November. The monthly mean minimum temperature is 19°C in January and 24°C in May.

***4. Cauvery Delta zone***

This zone comprises the Cauvery Delta area in Thanjavur, Thiruvarur, Nagapattinam districts and Musiri, Tiruchirapalli, Lalgudi, Thuraiyur and Kulithalai taluks of Tiruchirapalli district, Aranthangi taluk of Pudukottai district and Chidambaram and Kattumannarkoil taluks of Cuddalore district. The mean annual rainfall of the zone is 1078 mm out of which 40mm is received during winter, 69.2mm during summer, 295.4mm during South West Monsoon and 673.8mm during North East Monsoon.

***5. Southern zone***

This is the biggest among the seven zones of Tamil Nadu. It is typical zone surrounded by coastal areas on the East and mountains in the West. This zone comprises Sivagangai, Ramanathapuram, Virudunagar, Tuticorin and Tirunelveli districts and Natham and Dindigul taluks of Dindigul district, Melur, Tirumangalam, Madurai South and Madurai North taluks of Madurai district and Pudukkottai district excluding Aranthangi taluk. This zone lies on the Southern part of

the State under rain shadow area. Because of this, the area is prone to drought very often. The climate is semi-arid tropics. The elevation varies from mean sea level to 300 m. The mean annual rainfall is 776 mm received in 43 rainy days. The monthly mean maximum temperature in this region ranges from 28.5°C in December to 38.5°C in June. The monthly mean minimum temperature varies from 21.0°C in January to 27.5°C in June.

***6. High rainfall zone***

This zone consists of Kanayakumari district. This district situated in the southern most part of the Peninsular India, with its high rainfall having a climate which is entirely different from the rest of the state. The climate is monsoon tropics and there is seasonal in shores flow of moist air. The elevation ranges from sea level to about 600 m. The mean annual rainfall of the district is 1469 mm received in 64 rainy days. There is not much fluctuation in the mean monthly air temperature. The monthly mean maximum temperature varies from 28.0°C in December to 33.5°C in May. The monthly mean minimum temperature varies from 22°C in December to 26.5°C in May.

***7. High altitude and Hilly zone***

This zone comprises the hilly regions, namely the Nilgiris, Shevroys, Elagiri-Javvadhu, Kollimalai, Patchaimalai, Anamalais, Palanis and Podhigaimalais. The rainfall varies from 850 mm in Kalrayan hills to about 4500 mm in Anamalai hills.

**Agro-ecological zones of India**

An ecological region is characterized by distinct ecological responses to macroclimate as expressed in vegetation and reflected in soils, fauna and aquatic systems. Therefore, an agro-ecological region is the land unit on the earth's surface carved out of agro-climatic region when superimposed on different landform and soil conditions that act as modifiers of climate and length of growing period (LGP).

National Bureau of Soil Survey and Land Use Planning (NBSS & LUP) of the ICAR has delineated 20 agro-ecological regions (AERs) in the country using the FAO 1978 concept of superimposition of length of growing periods and bio-climate maps on soil physiographic map.

***Arid ecosystem***

1. Western Himalayas, cold eco-region, shallow soils, LGP <90 days.
2. Western plain Kachohh and parts of Kathiawar Peninsula, hot arid eco-region, desert and saline soils. LPG <90 days.
3. Deccan plateau, hot arid ecoregion, red and black soils. LGP <90 days.

***Semiarid ecosystem***

4. Northern plain and central high lands, hot semiarid eco-region, alluvial soils. LGP 90-150 days.
5. Central high lands, Gujarat plains, Kathiawar peninsula, hot semiarid eco-region, medium and deep black soils. LGP 90-150 days.
6. Deccan plateau, hot semiarid ecoregion. LGP 90-150 days.
7. Telangana, Eastern ghats, hot semiarid eco-region. LGP 90- 150 days.
8. Eastern ghats, Tamil Nadu uplands and Karanataka plateau, hot semiarid eco-region. LGP 90-150 days.

***Subhumid Ecosystem***

9. Northern plain, hot sub-humid (dry) eco-region, red and black soils. LGP 150-180 days.
10. Central highlands, hot sub-humid eco-region, black and red soils. LGP 150-180 (210) days.
11. Eastern plateau, hot sub-humid eco-region, red and yellow soils, (210) days. LGP 150-180 days.
12. Eastern plateau (Chotanagpur) and Eastern ghats hot sub-humid eco-region, red and lateritic soils. LGP 150-180 (210) days.
13. Eastern plain, hot sub-humid (moist) eco-region, alluvial soils. LGP 180-210 days.
14. Western Himalayas, warm sub-humid to humid eco-region with brown forest soils. LGP 180-210+ days.

***Humid-Perhumid ecosystem***

15. Bengal and Assam plain hot sub-humid (moist) to humid eco-region, alluvial soils. LGP 210+ days.
16. Eastern Himalayas, warm per-humid eco-region, brown and red hill soils. LGP 210 + days.
17. North eastern hills, warm per-humid eco-region, red and lateritic soils. LGP 210+ days.

***Coastal Ecosystem***

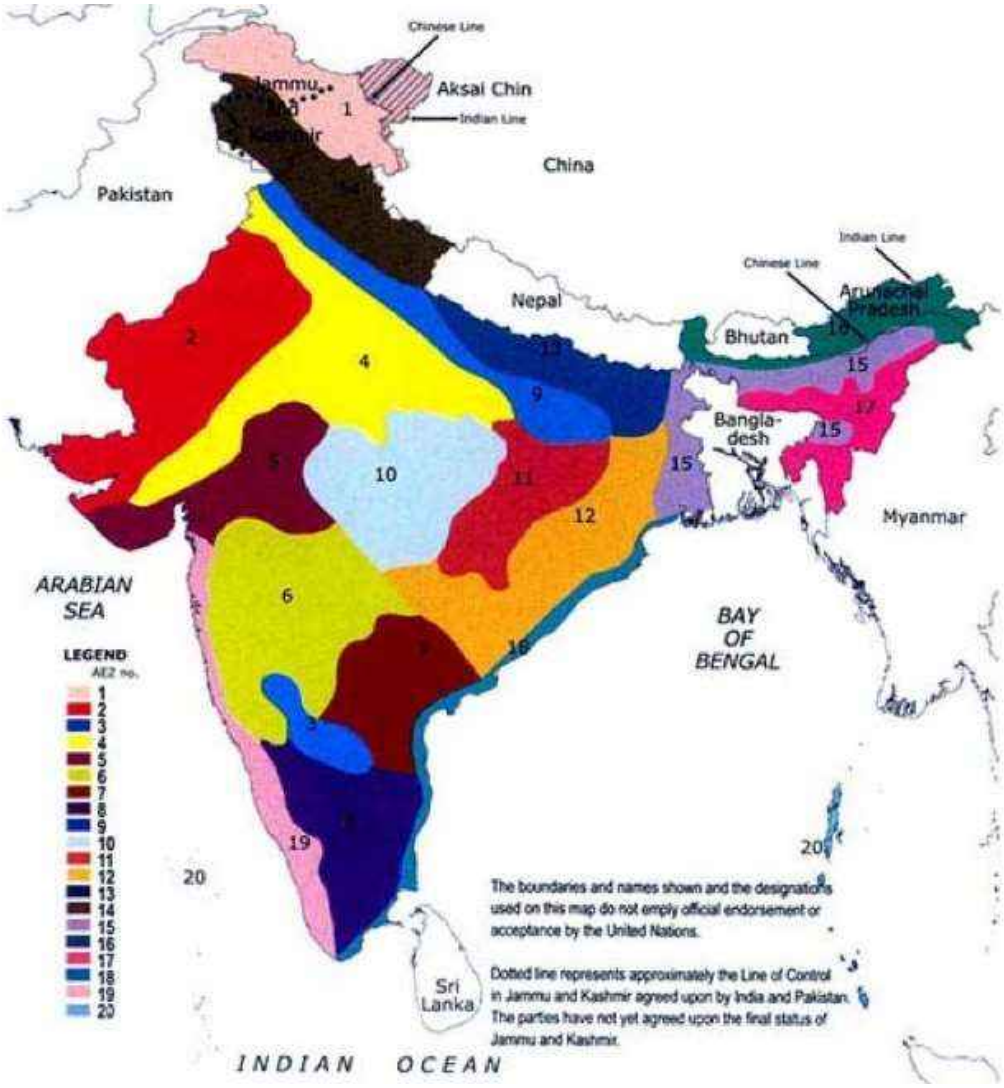
18. Eastern coastal plain, hot sub-humid to semiarid eco-region, coastal alluvium. LGP 90-210+ days.
19. Western Ghats and coastal plain, hot humid-per-humid eco-region, red, lateritic and alluvium derived soils. LGP 210+ days.

***Island Ecosystem***

20. Andaman Nicobar and Lakshadeep, hot humid to per-humid eco-region, real loamy and sandy soils. LGP 210+ days.

The major advantage of LGP based criteria is that the LGP is the direct indicative of moisture availability of a given landform rather than the total rainfall. For example, both Ratnagiri in western

Maharashtra and Nagpur in eastern Maharashtra have LGP 180-210 + days but the total annual rainfall of Ratnagiri is more than 2000 mm where as that of Nagpur is only 1100 mm. Therefore, agro-ecosystems approach allows crop planning based on length of growing period rather than the quantity of rainfall.



**Lec. 9. Crops and major soils - Classification – Economic and agricultural importance in India and Tamil Nadu**

## CROPS

In general, crop is an organism grown and / or harvested for obtaining yield. Agronomically, crop is a plant cultivated for economic purpose.

**Classification of crops**

Classification is done to generalize similar crop plants as a class for better understanding of them.

Classification is done to generalize similar crop plants as a class for attaining better understanding of them. Field crops are classified in the following ways.

1. According to range of cultivation
2. According to the place of origin
3. Botanical classification
4. Commercial classification
5. Economic / Agricultural / Agrarian classification
6. Seasonal classification
7. Classification based on ontogeny
8. According to cultural requirement
9. According to important uses

**1. Range of Cultivation**

a) **Garden crop** : Grown on a small scale in gardens. Example – Onion, Brinjal etc.,

b) **Plantation crop** : Grown on a large scale in estates and perennial in nature.

Example – Tea, Coffee, Cacao, Rubber etc.

c) **Field crop** : Grown on a vast scale under field condition. They are mostly seasonal such as rice, wheat, cotton etc. Agronomy deals with field crops only.

**2. Place of Origin**

**a) Native**: Crops grown within the geographical limits of their origin, for example – rice, barely, blackgram, green gram, mustard, castor, sugarcane and cotton, grown in India, are native to India.

b**) Exotic or Introduced**: Crops introduced from other countries, such as Tobacco, Potato, jute, maize, apple, etc.,

**3. Botanical / Taxonomical Classification** : According to systematic botany plants are classified as order, family etc., Similarly crop plants are grouped into families as,

a) Poaceae (Graminae) : Cereals, millets and grasses.

b) Papilionaceae (Legumes) : Pulses, legume fodders, vegetables, groundnut, berseem, green manures etc.

c) Cruciferae : Mustard, Indian rape seed, radish cabbage, cauliflower

d) Cucurbitaceae : All gourds, cucumber, pumpkin etc.

e) Malvaceae : Cotton, lady's finger, rosselle

f) Solanaceae : Potato, tomato, tobacco, chillies, brinjal

g) Tiliaceae : Jute

h) Asteraceae (Compositae) : Sunflower, safflower, niger,

i) Chenopodiaceae : Spinach, beet, sugarbeet.

j) Pedeliaceae : Sesame

k) Euphorbiaceae : Castor, tapioca

l) Convolvulaceae : Sweet potato

m) Umbelliferae : Coriander, cumin, carrot, anise

n) Liliaceae : Onion, garlic

o) Zingiberaceae : Ginger, turmeric

**4. Commercial classification** : Based on the plant products which come into the commercial field are grouped as

a) **Food crops** : Rice, wheat, greengram, soybean, groundnut, etc.

b) **Food crops / Forage crops** – All fodders, oats, sorghum, maize, napier grass, stylo, Lucerne etc.

c) **Industrial / Commercial crops** : Cotton, sugarcane, sugarbeet, tobacco, jute, etc.

d) **Food adjuncts** : Turmeric, garlic, cumin, etc.

**5. Economic / Agrarian / Agricultural Classification :** This classification is based on use of crop plants and their products. This is an important classification as for as agronomy is concerned (*Agronomic classification)*

**a) Cereals**: They are cultivated grasses grown for their edible starchy grains (one seeded fruit – caryopsis) Larger grains used as staple food are cereals – rice, wheat,

maize, barley, oats etc. The word cereal was derived from the word ceres which denotes a goddess who was believed as the giver of grains by Romans.

**b) Millets:** Small grained cereals which form the staple food in drier regions of the developing countries are called millets. Example – sorghum, bajra, ragi and minor millets.

**c) Oil seeds**: Crops that yield seeds that are rich in fatty acids, are used to extract vegetable oils. Example – mustard, rape seed, sesame, sunflower, castor, linseed, groundnut, soybeans etc.

**d) Pulses**: Seeds of leguminous plants used as food. They produce dal rich in protein. Examples – greengram, blackgram, redgram, bengal gram, soybean, cowpea, peas, lentil, lathyrus.

**e) Feed / Forage** : It refers to vegetative matter, fresh or preserved, utilized as feed for animals. It includes hay, silage, soilage, pasturage and fodder. Example – bajra napier grass, guinea grass, fodder sorghum, fodder maize, lucerne, desmanthus, etc.

**f) Fibre crops** : Plants grown for their fibre yield. There are different kinds of fibre. They are i) seed fibre – cotton, ii) stem/bast fibre-jute, mesta, iii) leaf fibre – agave, pineapple.

**g) Sugar and starch crops** : Crops grown for production of sugar and starch.

Example – sugarcane, sugarbeet, potato, sweet potato, tapioca and asparagus.

**h) Spices and Condiments** : Crop plants or their products used to season, flavour, taste, and add colour to the fresh or preserved food. Example ginger, garlic, fenugreek, cumin, turmeric, chillies, onion, coriander, anise and asafetida.

**i) Drug crops / Medicinal plants** : Crops used for preparation of medicines.

Example – tobacco, mint etc.

**j) Narcotics, fumitories and masticatories** : Plants / products used for stimulating, numbing, drowsing or relishing effects. Example – tobacco, ganja, opium poppy.

**k) Beverages** : Products of crops used for preparation of mild, agreeable and stimulating drinking. Example – tea, coffee, cocoa.

**6. Seasonal Classification** : Crops are grouped under the seasons in which their major field duration falls.

**a) Kharif or South West Monsoon season crops :** Crops grown during June – July to September – October which require a warm wet weather during their major period of growth and shorter day length for flowering. Example – rice, maize, castor, groundnut,

**b) Rabi Crops :** Crops grown during October – November to January – February which require cold dry weather for their major growth period and longer day length for flowering. Example – wheat, mustard, barley, oats, potato, bengal gram, berseem, cabbage and cauliflower.

**c) Zaid or summer crops** : Crops grown during February – March to May – June which require warm dry weather for growth and longer day – length for flowering. Example – blackgram, greengram, sesame, cowpea etc.

This classification is not a universal one. It only indicates the period when a particular crop is raised. Example – khairf rice, kharif maize, rabi, maize, summer pulse etc.

**7. According to Ontogeny :** It is a classification based on the life cycle of a plant.

**a) Annual Crops** : Crop plants that complete life cycle within a season for year. They produce seed and die within the season. Example-wheat, rice, maize, mustard.

**b) Biennial Crops** : Plants that have life span of two consecutive seasons or years. First year / Season these plants have purely vegetative growth usually confined to rosette of leaves. The tap root is often fleshy and serves as a food storage organ. During the second year / season they produce flower stocks from the crown and after producing seeds the plants die Example-sugarbeet, beet root, cabbage, radish, carrot, etc.

**c) Perennial Crops** : They live for three or more years. They may be seed bearing or non-seed bearing. Example – sugarcane, napier grass. In general perennial crops occupy land for more than 30 months.

**8. According to cultural requirement of crops**

Certain groups of plants are alike in cultural requirements due to their similar agrobotanical or morpho – agronomical characters.

**a)** ***According to suitability of toposequence***

**i) Crops grown on high land (dry land crops)** : Crops that cannot tolerant water stagnation come under this group example redgram, groundnut, maize, sorghum, cotton, sesame, napier etc.

**ii) crops grown on medium land (garden land crops) :** Crops that require sufficient soil moisture but cannot tolerate water stagnation. Example – Potato,sugarcane, upland rice, ragi, wheat, blackgram, bengal gram.

**iii) Crops grown on low land (Wetland crops)** : Crops that require abundant supply of water and can withstand prolonged water logged conditions. Example – rice, daincha, paragrass and jute.

**b)** ***According to the suitability of the textural groups of soils.***

**i) Crops suitable to sandy to sandyloam (light) soils** : Sorghum, bajra, green gram, sunflower, potato, onion, carrot etc.

**ii) Crops suitable to silty loam (medium) soils** : Jute, sugarcane, maize, cotton, mustard, tobacco, bengal gram, redgram, cowpea, etc.

**iii) Crop suitable to clay to clay loam (heavy) soils** : Rice, wheat, barley, linseed, lentil, paragrass, guinea grass, marvel grass etc.

**c)** ***According to tolerance to problem soils,***

i) Tolerant to acidic soils : Wet rice, potato, mustard.

ii) Tolerant to saline soils : Chillies, cucurbits, wheat, sorghum, bajra, cluster beans, barley etc.

iii) Tolerant to alkali / sodic soils : Barley, cotton, bengalgram, berseem, sunflower, maize, etc.

iv) Tolerant to waterlogged soils : Wet rice, daincha, paragrass, napier grass, guinea grass.

v) Crops tolerant to soil erosion : Marvel grass, groundnut, blackgram, rice bean, moth bean, horsegram.

**d)** ***According to tillage requirement***

i) Arable crops – require preparatory tillage. Example – Potato, tobacco, rice, maize.

ii) Non-arable crops – may not require preparatory cultivation / tillage. Example – Blackgram, green gram, Paragrass, cowpea.

**e)** ***According to the depth of root system***

i) Shallow rooted crops : Rice, potato, onion

ii) Moderately deep rooted : Wheat, groundnut, castor, tobacco.

iii) Deep rooted : Maize, cotton, sorghum

iv) Very deep rooted : Sugarcane, safflower, lucerne, redgram.

**f)** ***According to the tolerance to hazardous weather condition***

i) Frost tolerant : Sugarbeet, beet root,

ii) Cold tolerant : Potato, cabbage, mustard

iii) Drought tolerant : Bajra, jowar, barely, safflower, castor

**g)** ***According to water supply***

i) Irrigated crops : Rice, wheat, pulses, groundnut, berseem.

ii) Rainfed / uplant crops : Jute, maize, ragi, upland rice, redgram

iii) Rainfed but partially irrigated : Bengal gram, wheat, jowar, bajra, mustard

iv) Residual / conserved soil moisture : Rapeseed, bajra, barley, safflower, linseed.

v) Rainfed with supplemental irrigation : Kharif rice, sugarcane, blackgram.

vi) Rainfed with flooded water : Deep water rice, jute, sugarcane, daincha.

**h)** ***According to method of sowing / planting***

i) Direct seeded crop  Upland rice, wheat, jowar, bajra, groundnut etc.

ii) Planted crops : Sugarcane, potato, sweet potato, napier, guinea grass.

iii) Transplanted crops: (after raising seedlings in the nursery) : Rice, ragi, bajra, tobacco, bellary onion, brinjal.

***i) According to inter-tillage requirements specially earthing up***

i) Intertilled crops : Potato, sweet potato, groundnut, maize, sugarcane, turmeric, etc.

ii) Non-intertilled crops : Fodder sorghum, deenanath grass, paragrass etc.

j) *According to length of field duration of crops*

i) Very short duration crops : upto 75 days : Pulses

ii) Short duration crops : 75 – 100 days : Sunflower, cauliflower, uplandrice

iii) Medium duration crops : 100-125 days : Wheat, jowar, bajra, groundnut, sesame, jute.

iv) Long duration crops : 125-150 days : Mustard, tobacco, cotton

v) Very long duration crops : above 150 days : Sugarcane, redgram, castor.

**k)** ***According to the method of harvesting***

i) Reaping : Rice, wheat,

ii) Uprooting by pulling : Bengal gram, black gram, lentil, rapeseed

iii) Uprooting by digging : Potato, sweet potato, groundnut, carrot etc.

iv) Picking : Cotton, vegetables, brinjal, bhendi, chillies

v) Priming : Tobacco

vi) Cutting : Berseem, napier, amaranthus

vii) Grazing : Paragrass, kolukkattai grass, stylosanthes.

**l)** ***According to post harvest requirement***

i) Curing : Tobacco, mustard

ii) Stripping : Jute, sunnhemp

iii) Shelling : Groundnut

iv) Ginning : Cotton

v) Seasoning : Turmeric, chillies

vi) Grading and sorting : Potato, rice, wheat, fibre crops etc.

**m)** ***Based on crops growing soil condition***

i) Psammophytes (Sandy soil) : Castor

ii) Lithophytes (Rock surface) : Ferns.

iii) Chasmophytes (Rock crack) : Potato

iv) Acedophyes (Acid soil) : Potato

v) Basphytes (Alkali soil) : Rice

vi) Calciphytes (Basic soil) : Asparagus

vii) Halophytes (Saline soil) : Sugar beet, Alfalfa

**n)** ***Based on climatic condition***

1) Tropical crop : Coconut, sugarcane

2) Sub-tropical crop : Rice, cotton

3) Temperate crop : Wheat, barley

4) Polar crop : All pines, pasture grasses

**9. According to important uses:**

Though plants are useful in many ways only certain uses are given below.

**a) Catch crops / contingent crops :** are those crops cultivated to catch the forth coming season. It replaces the main crop that has failed due to biotic or climatic or management hazards. Generally, they are of very short duration, quick growing, fast bulking, harvestable or usable at any time of their field duration and adaptable to the season, soil and management practices. They provide feed, check weed growth, conserve soil, utilized added fertilizer and moisture Example – green gram, black gram, cowpea, onion, coriander, bajra.

**b) Restorative crops :** are those crops which provide a good yield along with enrichment or restoration of soil fertility or amelioration of the soils. They fix atmospheric nitrogen in root nodules, shed their leaves during ripening and thus restore soil conditions. Example-legumes.

**c) Exhaustive crops :** are those crop plants which on growing leave the field exhausted because of a more aggressive nature. Example-gingelly, brinjal, linseed, sunflower etc.

**d) Paira crop / residual crops :** are those crop plants which are sown a few days or weeks before the harvest of the standing mature crops to utilize the residual moisture, without preparatory tillage. The standing crop and the later sown (paira) crop become simultaneous (forming a pair) for a short period. For example Black gram in thaladi paddy; lathyrus, lentil etc., in paddy, Paira crops in succession may constitute relay cropping.

**e) Smother crops :** are those crop plants which are able to smother or suppress the weed growth by providing suffocation (curtailing movement of air) and obscuration (of the incidental radiation) through their dense foliage developed due to quick growing ability with heavy tillering or branching, planophyllic or procumbent or trailing habits. Example – barley, mustard, cowpea, etc.

**f) Cover crops :** are those crop plants which are able to protect the soil surface from erosion (wind, water or both) through their ground covering foliage and or root mats. Example – groundnut, blackgram, marvel grass, sweet potato.

**g) Nurse crops :** A companion crop which nourishes the main crop by way of nitrogen fixation and or adding the organic matter into the soil. Example-cowpea intercropped with cereals, glyricidia, tephrosia in tea.

**h) Guard / barrier crops:** are those crop plants which help to protect another crop from trespassing or restrict the speed of wind and thus prevent crop damage. Main crop in the centre surrounded by hardy or thorny crop. Example-mesta around sugarcane; sorghum around cotton; safflower around gram.

**i) Trap crops:** are those crop plants grown to trap soil borne harmful parasitic weeds.For example orobanche and striga are trapped by solanaceous and sorghum crops respectively. Nematodes are

trapped by solanaceous crops (On uprooting crop plants, nemathodes are removed from the soil). Castor in cotton, groundnut act as crop for army worm pest.

**j) Augmenting crops:** are those sub crops sown to supplement the yield of the main crop. Example – Mustard or cabbage with berseem to augument the forage yield of berseem.

**k) Alley crops :** are those arable crops which are grown in 'alleys' formed by trees or shrubs, established mainly to hasten soil fertility restoration, enhance soil productivity and reduce soil erosion. They are generally of non-trailing with shade tolerance capacity. For example growing pulses in between the rows of casuarina.

## SOILS

Soil is defined as the thin layer of earth's crust made up of disintegrated and decomposed rocks, complex mineral compound, organic matter, water/air and living organism like bacteria, fungi, insects and worms and serves as the natural medium of growth of plants.

It provides nutrients, moisture, anchorage (support) and provides air to root system. There are different soil groups found in varied regions of India. Each group differs from other in physical and chemical properties. The variation in behaviour is mainly due to the nature of the parent material from which the soils are formed. Parent materials are Igneous rocks, sedimentary rocks and metamorphic rocks. Physical properties like structure, texture, colour, water holding capacity, depth etc. are to be noted. Chemical properties like the presence of various plant elements, pH, EC, CEC, acidic or alkaline, etc. are considered.

**Classification based on soil taxonomy**

| | | | |
|---|---|---|---|
| Order | : *Entisols* | Suborder | : Fluvents |
| Great Group | : Torrifluvents | Subgroup | : Typic Torrifluvents |
| Family | : Fine-loamy, mixed, superactive, calcareous, Typic Torrifluvents | | |
| Series | : Jocity, Youngston. | | |

**Major soils of India**

1. Alluvial soil (*Entisols, Inceptisols and Alfisol)*
2. Black soil (*Vertisol*)
3. Red soil (*Alfisol*)
4. Laterite soil (*Ultisol*)
5. Desert soil (*Aridisol*)
6. Forest soil and hill soil, peat and marshy soils
7. Problem soils (saline, alkali, acid)

**1. Alluvial soil or Indo-Gangetic Alluvium**

This is the most extensive soil found in India. Out of total area of India, 48.0 m.ha comes under river alluvium. These soils include deltaic alluvium, calcareous alluvium and coastal alluvium. Alluvial soils are formed by transportation in streams and rivers and are deposited in flood plains or along the coastal belts. Newly formed alluvium may not have distinct soil horizons while older alluvium may have soil horizons. They occur in the basins of Indus, Ganges, Brahmaputra, Godavari, Krishna, Cauvery and Tambiraparani deltas spread in U.P., Bihar, West Bengal, Gujarat, Punjab, Rajasthan, Andhra Predesh, Tamil Nadu.

Newer alluvium is called as *Khadar*, is sandy, light colour and less *Kankar* nodules. Older alluvium is called as *Bhangar*, full of clay, dark colour and more *Kankar* nodules. Alluvial soils of high altitude are acidic in nature and plains are neutral to alkaline. Alluvial soils of plains are medium in phosphorous content and high in potassium content. Generally, alluvial soils are rich in nutrients and are fertile and they support good crop growth with plenty of water. Many crops including vegetable are cultivated in river alluvium. Crops like rice, wheat, cotton, maize, sugarcane, vegetables, jute, oil seeds, millets, pulses and fruits are cultivated in these soil.

**2. Black soil**

Dark-grey in colour due to clay-humus complex. Area around 32.0 m.ha is under this soil. This soil is also called black cotton soil, mixing of soil along the entire column with *Montmorillonite* clay. Cotton grows very well with water available in soil. Black soil holds more moisture and available for a long time. Found in Maharashtra, Madhya Pradesh, South Orissa, South and Coastal Andhra Pradesh, North Karnataka and parts of Tamil Nadu. Black soil contains high proportion of clay (30-40%), so, the water holding capacity is high. Typical characteristics of this black soil are swelling (during wet period) and shrinkage (dry period). While dry, it forms very deep cracks of more than 30-45 cm. In Kovilpatti (Tamil Nadu) areas the cracks may extend to 2 to 3 m with a width of 1 to 6 cm. Field preparation takes longer time compared to other soil. Only after secondary tillage, the soil is suited for crop production. The soils are fine grained contain high proportion of Calcium and Magnesium carbonates. They are poor in N, medium in P and medium to high in K (Characteristic feature of typical Indian soil).

In Tamil Nadu Black soils have high pH (8.5 to 9) and are rich in lime (5-7%), have low permeability. The soils are with more cation exchange capacity (40-60 m.e./100 g). Crops grown in this soil are cotton, bengal gram, mustard, millets, pulses, oil seeds (sunflower, safflower) are commonly grown in this soil. Most of the soils come under rainfed areas.

**3. Red soil**

Based on the colour (due to presence of ferric oxides) it is called as red soil. Around 30 m.ha are found in India. They are formed from granites and other metamorphic rocks. Mostly found in semi-arid areas and the colour varies from red to yellow. The soil is light textured, with *Kaolinite* type of clay. Well drained with moderate permeability. Low cation exchange capacity and low water holding capacity. Red soil is present in Gujarat, Tamil Nadu, Karnataka, Andhra Pradesh, North and East of Arunachal Pradesh, Madhya Pradesh, Parts of Bihar and Uttar Pradesh. They are shallow in depth because they are degraded or drained soil. Lesser clay and more sandy than *Vertisol*. Red soil is always in acidic nature. Highly suitable for groundnut crop cultivation. Crops like millets, pulses, oil seeds (ground nut, gingelly, castor) and tuber crops like cassava are commonly cultivated.

**4. Laterites and Lateritic soil**

Laterite soils are formed due to the process of laterisation. i.e., leaching of all cations leaving Fe and Al oxides. Mostly found in hills and foothill areas. This soil is formed under high intensive down pour of rainfall. It is modified form of red soil, clay content is minimum. Rich in organic matter content and rich in fertility and medium water holding capacity. They become very hard when there is no water. The cohesive nature is high. Acid loving crops (Plantation crops) and fruits (pineapple, avacado) are more cultivated. Tea, rubber, pepper, spices are cultivated. At lower elevation places, rice is grown.

**5. Desert soil**

Found in desert regions of Rajasthan (Thar desert), parts of Haryana and Punjab of India. More sand is found and sand dunes are common. Clay content is < 8% only. Poor fertility, poor water holding capacity and susceptible to soil erosion. Presence of sodic salts (high Na content) leads to alkalinity. Crops like date palm, cucumber, millets are cultivated (countries like Saudi Arabia, UAE, Jordan, Sudan etc).

**6. Peaty and Organic soil**

These soils are very rich in organic matter. Found in Kerala, coastal regions of West Bengal, Orissa, South and East coast of Tamil Nadu. Deposition of organic matter by the elevated soil. Peaty and organic soil is not suitable for majority of crops. Rice is mostly cultivated in coastal area in rainy season.

**7. Problem soil**

***Saline soils:*** Contain excess amounts of neutral soluble salts dominated by chlorides and sulphates of Na, Ca and Mg affects plant growth. White encrustation of salts and hence called white alkali.

These soils are characterized by, EC: 4dSm$^{-1}$ at 25°C, ESP: < 15; pH; < 8.5. This soil needs leaching and drainage before cropping for amelioration.

i) High salt tolerant: *Sesbania*, Rice, sugarcane, oats, berseem, lucerne, indian clover & barley.

ii) Medium salt tolerant: Castor, cotton, sorghum, pearl millet, maize, mustard & wheat.

iii) Low salt tolerant: Pulses, peas, *Sunnhemp*, gram, linseed and sesame.

***Sodic / Alkali soils:*** High content of carbonates and bicarbonates of Na. Hence, they are with high exchangeable sodium percentage (ESP) with dark encrustation, hence called as black alkali. These soils are rich in $NaHCO_3$ and characterized by pH: > 8.5; EC : < 4dSm$^{-1}$; ESP : > 15. Use gypsum ($CaSO_4$, $2H_2O$) as amendment for reclamation of sodic alkali soils. Iron pyrites ($FeS_2$), bulky organic manures (especially green manure) and crop residues which produces weak organic acids.

i) Tolerant crops: Karnal / rhodes / para/ bermuda grass, rice and sugar beet.

ii) Semi–tolerant: Wheat, barley, oats, berseem and sugarcane.

iii) Sensitive: Cowpea, gram, groundnut, lentil, peas and maize.

***Acid soils:*** These are low pH with high amounts of exchangeable $H^+$ and $Al_3$. Occur in regions with high rainfall. Significant amount of partly decomposed organic matter exist. Have low CEC and high base saturation. Liming and judicious use of fertilizers are the management measures suggested. Suitable crops: Acedophytes (like potato).

**Comparison of three types of soils**

| Parameters | Saline soil | Saline alkali | Alkali soil |
|---|---|---|---|
| EC (dS/m) | >4 | >4 | <4 |
| ESP (%) | <15 | >15 | >15 |
| pH | <8.5 | <8.5 | >8.5 |

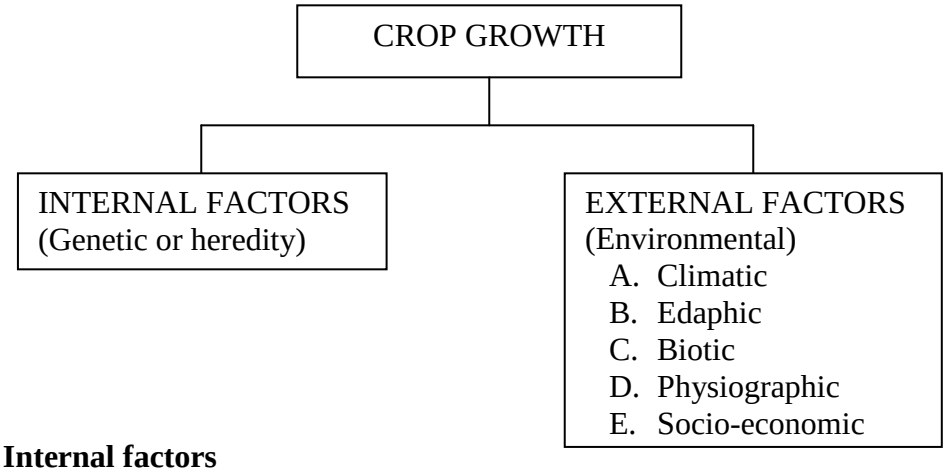
**Soils of Tamil Nadu**

| Type of Soil | Areas in Tamil Nadu |
|---|---|
| Red loam<br>(79.8 L. ha & 61.7%) | Parts of Kancheepuram, Cuddalore, Salem, Dharmapuri, Coimbatore, Tiruchirappalli, Thanjavur, Ramanathapuram, Madurai, Tirunelveli, Sivagangai, Thoothukudi, Virudhunagar, Dindigul and The Nilgiris Districts. |
| Laterite soil<br>(3.8 L.ha & 2.9%) | Parts of The Nilgiris District |
| Black soil<br>(15.0 L. ha & 11.6%) | Parts of Kancheepuram, Cuddalore, Vellore, Thiruvannamalai, Salem, Dharmapuri, Madurai, Ramanathapuram, Tirunelveli, Sivagangai, Thoothukudi, The Nilgiris, Virudhunagar and Dindigul Districts. |
| Sandy coastal alluvium | On the Coasts in the districts of Ramanathapuram, Thanjavur, Nagapattinam, |

| | |
|---|---|
| (9.8 L. ha & 7.6%) | Cuddalore, Kancheepuram and Kanyakumari |
| River alluvium<br>(21.0 L. ha & 16.2%) | All river deltaic areas (Cauvery, Vaigai, Tambiraparani) |

**Source:** Department of Economics and Statistics, Chennai

**Lect 10. Factors affecting crop production – climatic – edaphic - biotic- physiographic and socio economic factors**




**I. Internal factors**

**Genetic factors**

The increase in crop yields and other desirable characters are related to Genetic make up of plants.

- High yielding ability
- Early maturity
- Resistance to lodging
- Drought flood and salinity tolerance
- Tolerance to insect pests and diseases
- Chemical composition of grains (oil content, protein content )
- Quality of grains (fineness, coarseness)
- Quality of straw (sweetness, juiciness)

The above characters are less influenced by environmental factors since they are governed by genetic make-up of crop.

**2. External factors**

A. Climatic
B. Edaphic
C. Biotic
D. Phsiographic
E. Socio-economic

## A. CLIMATIC FACTORS

Nearly 50 % of yield is attributed to the influence of climatic factors. The following are the atmospheric weather variables which influences the crop production.

1. Precipitation
2. Temperature
3. Atmospheric humidity
4. Solar radiation
5. Wind velocity
6. Atmospheric gases

### 1. Precipitation

- Precipitation includes all water which falls from atmosphere such as rainfall, snow, hail, fog and dew.
- Rainfall one of the most important factor influences the vegetation of a place.
- Total precipitation in amount and distribution greatly affects the choice of a cultivated species in a place.
- In heavy and evenly distributed rainfall areas, crops like rice in plains and tea, coffee and rubber in Western Ghats are grown.
- Low and uneven distribution of rainfall is common in dryland farming where drought resistance crops like pearl millet, sorghum and minor millets are grown.
- In desert areas grasses and shrubs are common where hot desert climate exists
- Though the rainfall has major influence on yield of crops, yields are not always directly proportional to the amount of Precipitation as excess above optimum reduces the yields
- Distribution of rainfall is more important than total rainfall to have longer growing period especially in drylands

### 2. Temperature

- Temperature is a measure of intensity of heat energy. The range of temperature for maximum growth of most of the agricultural plants is between 15 and 40ºC.
- The temperature of a place is largely determined by its distance from the equator (latitude) and altitude.
- It influences distribution of crop plants and vegetation.
- Germination, growth and development of crops are highly influenced by temperature.
- Affects leaf production, expansion and flowering.

- Physical and chemical processes within the plants are governed by air temperature.
- Diffusion rates of gases and liquids changes with temperature.
- Solubility of different substances in plant is dependent on temperature.
- The minimum, maximum (above which crop growth ceases) and optimum temperature of individual's plant is called as cardinal temperature.
-

| Crops | Minimum temperature ºC | Optimum temperature ºC | Maximum temperature ºC |
|---|---|---|---|
| Rice | 10 | 32 | 36-38 |
| wheat | 4.5 | 20 | 30-32 |
| Maize | 8-10 | 20 | 40-43 |
| Sorghum | 12-13 | 25 | 40 |
| Tobacco | 12-14 | 29 | 35 |

**Cold injury**

Plants are injured due to very low temperature in the following wasy:

**1. Chilling injury** : Plants growing in hot climate, if exposed to low temperature (which is above the freezing point) for some time, are found to be killed or injured severely.

Example – Chlorotic condition or bands on leaves of sugarcane, sorghum and maize when exposed for 60 hours at 2 to $4_0$C.

**2. Freezing injury** : This is generally caused in plants growing in temperate region. When the plants are exposed to very low temperature, water is frozen into ice crystals inthe intercellular spaces. With further fall in temperature, water is withdrawn from the cells, resulting in increase in the size of ice crystals in the intercellular spaces. The protoplasm of the cell is dehydrated, and mechanical distortion takes place resulting in killing of the cells. Frost damage to potatoes, tea etc. in winter in the hilly areas like The Nilgiris is a typical example of the freezing injury.

**3. Suffocation** : During winter the ice or the snow form a thick cover over the ground and the crop suffers for want of oxygen. Ice in contact with roots inhibits diffusion of carbon dioxide and the respiratory products may become harmful to plants.

**4. Heaving** : Injury to plants is caused by a lifting upward of the plant along with the soil from its normal position in temperate regions where snowfall is common.

**Heat injury**

Very high temperature often stops growth. The plant faces incipient starvation due to high respiration rates. The plant is stunted and if such a condition persists for a long period the plant is killed. Direct temperature effects are noticeable in young seedlings and transplanted crops. High temperature causes sterility in flowers. The general effects of excessive heat are defoliation, premature dropping of fruits and in extreme cases death of plants.

**Effects of temperature on crop production**

Plants can grow only within certain limits of temperature. For each species and variety there are not only optimal temperature limits, but also optimal temperatures for different growth stages and functions, as well as lower and upper lethal limits. During photosynthesis there are certain biochemical processes preceding and following the reduction of carbon dioxide, which are affected mainly by temperature. As long as light is limiting, temperature has little effect on the rate of photosynthesis. When light is not limiting, it has a profound effect on the rate of photosynethesis.

In general, high temperature accelerates growth process. Rarely are high temperatures *per se* the direct cause of death of plants, provided the water supply is adequate.

Retardation of growth and difficulties in fertilization are observed even in heat loving crops such as sorghum, at extremes of temperatures. The harmful effects of excessive temperatures are usually aggravated by lack of available moisture. Hot dry winds will further increase the damage. Increasing temperatures increases evapotranspiration. The daily alternations of high and low temperature may effect grain production

**3. Atmospheric Humidity (Relative Humidity - RH)**

- Water is present in the atmosphere in the form of invisible water vapour, normally known as humidity. Relative humidity is ratio between the amount of moisture present in the air to the saturation capacity of the air at a particular temperature.
- If relative humidity is 100% it means that the entire space is filled with water and there is no soil evaporation and plant transpiration.
- Relative humidity influences the water requirement of crops
- Relative humidity of 40-60% is suitable for most of the crop plants.
- Very few crops can perform well when relative humidity is 80% and above.
- When relative humidity is high there is chance for the outbreak of pest and disease.

**4. Solar radiation (without which life will not exist)**

- From germination to harvest and even post harvest crops are affected by solar radiation.
- Biomass production by photosynthetic processes requires light.

- All physical process taking place in the soil, plant and environment are dependent on light
- Solar radiation controls distribution of temperature and there by distribution of crops in a region.
- Visible radiation is very important in photosynthetic mechanism of plants. Photosynthetically Active Radiation (PAR - 0.4 – 0.7μ) is essential for production of carbohydrates and ultimately biomass.

  0.4 to 0.5 μ - Blue – violet – Active

  0.5 to 0.6 μ - Orange – red - Active

  0.5 to 0.6 μ - Green –yellow – low active
- Photoperiodism is a response of plant to day length

  Short day – Day length is <12 hours (Rice, Sunflower and cotton), long day – Day length is > 12 hours (Barley, oat, carrot and cabbage), day neutral – There is no or less influence on day length (Tomato and maize).
- Phototropism — Response of plants to light direction. Eg. Sunflower
- Photosensitive – Season bound varieties depends on quantity of light received

**5. Wind velocity**

- The basic function of wind is to carry moisture (precipitation) and heat.
- The moving wind not only supplies moisture and heat, also supplies fresh $CO_2$ for the photosynthesis.
- Wind movement for 4 – 6 km/hour is suitable for more crops.
- When wind speed is enormous then there is mechanical damage of the crops (i.e.) it removes leaves and twigs and damages crops like banana, sugarcane
- Wind dispersal of pollen and seeds is natural and necessary for certain crops.
- Causes soil erosion.
- Helps in cleaning produce to farmers.
- Increases evaporation.
- Spread of pest and diseases.

**6. Atmospheric gases on plant growth**

- $CO_2$ – 0.03%, $O_2$ - 20.95%, $N_2$ - 78.09%, Argon - 0.93%, Others - 0.02%.
- $CO_2$ is important for Photosynthesis, $CO_2$ taken by the plants by diffusion process from leaves through stomata

- $CO_2$ is returned to atmosphere during decomposition of organic materials, all farm wastes and by respiration
- $O_2$ is important for respiration of both plants and animals while it is released by plants during Photosynthesis
- Nitrogen is one of the important major plant nutrient, Atmospheric N is fixed in the soil by lightning, rainfall and N fixing microbes in pulses crops and available to plants
- Certain gases like $SO_2$, CO, $CH_4$, HF released to atmosphere are toxic to plants

**B. EDAPHIC FACTORS (soil)**

Plants grown in land completely depend on soil on which they grow. The soil factors that affect crop growth are

1. Soil moisture
2. Soil air
3. Soil temperature
4. Soil mineral matter
5. Soil organic matter
6. Soil organisms
7. Soil reactions

**1. Soil moisture**

- Water is a principal constituent of growing plant which it extracts from soil
- Water is essential for photosynthesis
- The moisture range between field capacity and permanent wilting point is available to plants.
- Available moisture will be more in clay soil than sandy soil
- Soil water helps in chemical and biological activities of soil including mineralization
- It influences the soil environment Eg. it moderates the soil temperature from extremes
- Nutrient availability and mobility increases with increase in soil moisture content.

**2. Soil air**

- Aeration of soil is absolutely essential for the absorption of water by roots
- Germination is inhibited in the absence of oxygen
- $O_2$ is required for respiration of roots and micro organisms.
- Soil air is essential for nutrient availability of the soil by breaking down insoluble mineral to soluble salts

- For proper decomposition of organic matter
- Potato, tobacco, cotton linseed, tea and legumes need higher $O_2$ in soil air
- Rice requires low level of $O_2$ and can tolerate water logged (absence of O2) condition.

**3. Soil temperature**

- It affects the physical and chemical processes going on in the soil.
- It influences the rate of absorption of water and solutes (nutrients)
- It affects the germination of seeds and growth rate of underground portions of the crops like tapioca, sweet potato.
- Soil temperature controls the microbial activity and processes involved in the nutrient availability
- Cold soils are not conducive for rapid growth of most of agricultural crops

**4. Soil mineral matter**

- The mineral content of soil is derived from the weathering of rocks and minerals as particles of different sizes.
- These are the sources of plant nutrients

  eg; Ca, Mg, S, Mn, Fe, K etc

**5. Soil Organic matter**

- It supplies all the major, minor and micro nutrients to crops
- It improves the texture of the soil
- It increases the water holding capacity of the soil,
- It is a source of food for most microorganisms
- Organic acids released during decomposition of organic matter enables mineralisation process thus releasing unavailable plant nutrients

**6. Soil organisms:**

- The raw organic matter in the soil is decomposed by different micro organisms which in turn releases the plant nutrients
- Atmospheric nitrogen is fixed by microbes in the soil and is available to crop plants through symbiotic (*Rhizobium*) or non-symbiotic (*Azospirillum*) association

**7. Soil reaction (pH)**

- Soil reaction is the pH (hydrogen ion concentration) of the soil.
- Soil pH affects crop growth and neutral soils with pH 7.0 are best for growth of most of the crops

- Soils may be acidic (<7.0), neutral (=7.0), saline and alkaline (>7.0)
- Soils with low pH is injurious to plants due high toxicity of Fe and Al.
- Low pH also interferes with availability of other plant nutrients.

**C. BIOTIC FACTORS**

Beneficial and harmful effects caused by other biological organism (plants and animals) on the crop plants

**1. Plants**

- Competitive and complimentary nature among field crops when grown together
- Competition between plants occurs when there is demand for nutrients, moisture and sunlight particularly when they are in short supply or when plants are closely spaced
- When different crops of cereals and legumes are grown together, mutual benefit results in higher yield (synergistic effect)
- Competition between weed and crop plants as parasites eg: *Striga* parasite weed on sugarcane crop

**2. Animals**

- Soil fauna like protozoa, nematode, snails, and insects help in organic matter decomposition, while using organic matter for their living
- Insects and nematodes cause damage to crop yield and considered as harmful organisms.
- Honey bees and wasps help in cross pollination and increases yield and considered as beneficial organisms
- Burrowing earthworm facilitates aeration and drainage of the soil as ingestion of organic and mineral matter by earthworm results in constant mixing of these materials in the soils.
- Large animals cause damage to crop plants by grazing (cattle, goats etc)

**D. Physiographic factors:**

- Topography is the nature of surface earth (leveled or sloppy) is known as topography. Topographic factors affect the crop growth indirectly.
- Altitude – increase in altitude cause a decrease in temperature and increase in precipitation and wind velocity (hills and plains)
- Steepness of slope: it results in run off of rain water and loss of nutrient rich top soil
- Exposure to light and wind: a mountain slope exposed to low intensity of light and strong dry winds may results in poor crop yields (coastal areas and interior pockets)

**E. Socio-economic factors**

- Society inclination to farming and members available for cultivation
- Appropriate choice of crops by human beings to satisfy the food and fodder requirement of farm household.
- Breeding varieties by human invention for increased yield or pest & disease resistance
- The economic condition of the farmers greatly decides the input/ resource mobilizing ability (marginal, small, medium and large farmers)

**Lec. 11. Tillage – Definition – objectives – types of tillage - modern concepts of tillage – main field preparation**

**Tillage**

Tillage operations in various forms have been practiced from the very inception of growing plants. Primitive man used tools to disturb the soils for placing the seeds. The word tillage is derived from 'Anglo-Saxon' words *Tilian* and *Teolian*, meaning 'to plough and prepare soil for seed to sow, to cultivate and to raise crops'. **Jethrotull,** who is considered as father of tillage suggested that thorough ploughing is necessary so as to make the soil into fine particles.

**Tillage** is the mechanical manipulation of soil with tools and implements for obtaining conditions ideal for seed germination, seedling establishment and growth of crops.

**Tilth** is the physical condition of soil obtained out of tillage (or) it is the result of tillage. The tilth may be a coarse tilth, fine tilth or moderate tilth.

**Objectives of tillage**

The main objectives of tillage are,

- To prepare a good seed bed which helps the germination of seeds.
- To create conditions in the soil suited for better growth of crops.
- To control the weeds effectively.
- To make the soil capable for absorbing more rain water.
- To mix up the manure and fertilizers uniformly in the soil.
- To aerate the soil.
- To provide adequate seed-soil contact to permit water flow to seed and seedling roots.
- To remove the hard pan and to increase the soil depth.

To achieve these objectives, the soil is disturbed / opened up and turned over.

**Types of tillage:** Tillage operations may be grouped into

1. On season tillage 2. Off-season tillage

**<u>1. On-season tillage</u>**

Tillage operations that are done for raising crops in the same season or at the onset of the crop season are known as on-season tillage. They may be preparatory cultivation and after cultivation.

***A. Preparatory tillage:*** This refers to tillage operations that are done to prepare the field for raising crops. It consists of deep opening and loosening of the soil to bring about a desirable tilth as well as to incorporate or uproot weeds and crop stubble when the soil is in a workable condition.

***Types of preparatory tillage***

a. Primary tillage
b. Secondary tillage

***a. Primary tillage:*** The tillage operation that is done after the harvest of crop to bring the land under cultivation is known as primary tillage or ploughing. Ploughing is the opening of compact soil with the help of different ploughs. Country plough, mould board plough, bose plough, tractor and power tiller drawn implements are used for primary tillage.

***b. Secondary tillage:*** The tillage operations that are performed on the soil after primary tillage to bring a good soil tilth are known as secondary tillage. Secondary tillage consists of lighter or finer operation which is done to clean the soil, break the clods and incorporate the manure and fertilizers. Harrowing and planking is done to serve those purposes.

*Planking* is done to crush the hard clods, level the soil surface and to compact the soil lightly. Harrows, cultivators, *Guntakas* and spade are used for secondary tillage.

***c. Layout of seed bed:***This is also one of the components of preparatory tillage. Leveling board, buck scrapers etc. are used for leveling and markers are used for layout of seedbed.

***B. After cultivation (Inter tillage):*** The tillage operations that are carried out in the standing crop after the sowing or planting and prior to the harvesting of the crop plants are called after tillage. This is also called as inter cultivation or post seeding/ planting cultivation. It includes harrowing, hoeing, weeding, earthing up, drilling or side dressing of fertilizers etc. Spade, hoe, weeders etc. are used for inter cultivation.

**2. Off-season tillage:** Tillage operations done for conditioning the soil suitably for the forthcoming main season crop are called off-season tillage. Off season tillage may be, post harvest tillage, summer tillage, winter tillage and fallow tillage.

**Special purpose tillage:** Tillage operations intended to serve special purposes are said to be special purpose tillage. They are,

**a. Sub-soiling:** To break the hard pan beneath the plough layer, special tillage operation (chiseling) is performed to reduce compaction. Sub-soiling is essential and once in four to five years where heavy machineries are used for field operations, seeding, harvesting and transporting. Advantages of sub-soiling are, greater volume of soil may be obtained for cultivation of crops, excess water may percolate downward to recharge the permanent water table, reduce runoff and soil erosion and roots of crop plants can penetrate deeper to extract moisture from the water table.

**b. Clean tillage:** It refers to working of the soil of the entire field in such a way no living plant is left undisturbed. It is practiced to control weeds, soil borne pathogen and pests.

**c. Blind tillage:** It refers to tillage done after seeding or planting the crop (in a sterile soil) either at the pre-emergence stage of the crop plants or while they are in the early stages of growth so that crop plants (sugarcane, potato etc.) do not get damaged, but, extra plants and broad leaved weeds are uprooted.

**d. Dry tillage:** Dry tillage is practiced for crops that are sown or planted in dry land condition having sufficient moisture for germination of seeds. This is suitable for crops like broadcasted rice, jute, wheat, oilseed crops, pulses, potato and vegetable crops. Dry tillage is done in a soil having sufficient moisture (21-23%). The soil becomes more porous and soft due to dry tillage. Besides, the water holding capacity of the soil and aeration are increased. These conditions are more favourable for soil micro-organisms.

**e. Wet tillage or puddling:** The tillage operation that is done in a land with standing water is called wet tillage or puddling. Puddling operation consists of ploughing repeatedly in standing water until the soil becomes soft and muddy. Puddling creates an impervious layer below the surface to reduce deep percolation losses of water and to provide soft seed bed for planting rice. Puddling is done in both the directions for the incorporation of green manures and weeds. Wet tillage destroys the soil structure and the soil particles that are separated during puddling settle later. Wet tillage is the only means of land preparation for transplanting semi-aquatic crop plant such as rice. Planking after wet tillage makes the soil level and compact. Puddling hastens transplanting operation as well as establishment of seedlings. Wet land ploughs or worn out dry land ploughs are normally used for wet tillage.

**Depth of ploughing**

The desirable depth of ploughing is 12 to 20 cm for field crops. The ploughing depth varies with effective root zone of the crop. The depth of ploughing is 10-20 cm for shallow rooted crops and 15-30 cm for deep rooted crops.

**Number of ploughing**

Number of ploughing depends on soil conditions, time available for cultivation between two crops and type of cropping systems. Zero tillage is practiced in rice fallow pulses. Minimum number of ploughing is taken up at optimum moisture level to bring favourable tilth depending on need of the crop.

**Time of ploughing**

The optimum soil moisture content for tillage is 60% of field capacity.

**Modern concepts in tillage:**

Conventional tillage involves primary tillage to break open and turn the soil followed by secondary tillage to obtain seed bed for sowing or planting. With the introduction of herbicides in intensive farming systems, the concept of tillage has been changed. Continuous use of heavy ploughs create hard pan in the subsoil, results in poor infiltration. It is more susceptible to run-off and erosion. It is capital intensive and increase soil degradation. To avoid these ill effects, modern concepts on tillage is in rule.

**1. Minimum tillage:** It aims at reducing tillage operations to the minimum necessity for ensuring a good seed bed. The advantages of minimum tillage over conventional tillage are,

- The cost and time for field preparation is reduced by reducing the number of field operations.
- Soil compaction is comparatively less.
- Soil structure is not destroyed.
- Water loss through runoff and erosion is minimum.
- Water storage in the plough layer is increased.

Tillage can be reduced in 2 ways

1. By omitting operations which do not give much benefit when compared to the cost.
2. By combining agricultural operations like seeding and fertilizer application.

The minimum tillage systems can be grouped into the following categories,

1. Row zone tillage

Primary tillage is done with mould board plough in the entire area of the field; secondary tillage operations like discing and harrowing are reduced and done only in row zone.

2. Plough plant tillage

After the primary tillage, a special planter is used for sowing. In one run over the field, the row zone is pulverized and seeds are sown by the planter

3. Wheel track tillage

Primary ploughing is done as usual. Tractor is used for sowing; the wheels of the tractor pulverize the row zone in which planting is done.

In all these systems, primary tillage is as usual. However, secondary tillage is replaced by direct sowing in which sown seed is covered in the row zone with the equipment used for sowing.

**2. Zero tillage (No tillage):** In this, new crop is planted in the residues of the previous crop without any prior soil tillage or seed bed preparation and it is possible when all the weeds are controlled by the use of herbicides. Zero tillage is applicable for soils with a coarse textured surface horizon, good

internal drainage, high biological activity of soil fauna, favourable initial soil structure and an adequate quantity of crop residue as mulch. These conditions are generally found in *Alfisols*, *Oxisols* and *Ultisols* in the humid and sub-humid tropics.

*Till planting*

Till planting is one method of practicing zero tillage. A wide sweep and trash bar clears a strip over the previous crop row and planter opens a narrow strip into which seeds are planted and covered. Here, herbicide functions are extended. Before sowing, the vegetation present has to be destroyed for which broad spectrum non selective herbicides like glyposate, paraquat and diquat are used.

Advantages

- Zero tilled soils are homogenous in structure with more number of earthworms.
- Organic matter content increases due to less mineralization.
- Surface run-off is reduced due to presence of mulch.

Disadvantages

- Higher amount of nitrogen has to be applied for mineralization of organic matter in zero tillage.
- Perennial weeds may be a problem.
- High number of volunteer plants and buildup of pests.

**3. Stubble mulch tillage or stubble mulch farming**

Soil is protected at all times either by growing a crop or by leaving the crop residues on the surface during fallow periods. Sweeps or blades are generally used to cut the soil up to 12 to 15 cm depth in the first operation after harvest and depth of cut is reduced during subsequent operations. When large amount of residues are present, a disc type implement is used for the first operation to incorporate some of the residues into the soil. This hastens the decomposition but still keeps enough residues on top soil.

Two methods for sowing crops in stubble mulch tillage are,

1. Similar to zero tillage, a wide sweep and trash bars are used to clear a strip and a narrow planter shoe opens a narrow furrow into which seeds are placed.
2. A narrow chisel of 5-10 cm width is worked through the soil at a depth of 15-30 cm leaving all plant residues on the surface. The chisel shatters the tillage pans and surface crusts. Planting is done with special planters.

Disadvantages of stubble mulch farming

- The residues left on the surface interfere with seed bed preparation and sowing operations.
- The traditional tillage and sowing implements or equipments are not suitable under these conditions.

**4. Conservation tillage:** The major objective is to conserve soil and soil moisture. It is a system of tillage in which organic residues are not inverted into the soil such that they remain on surface as protective cover against erosion and evaporation losses of soil moisture. If stubble forms the protective cover on the surface, it is usually referred to as stubble mulch tillage. The residues left on soil surface interfere with seed bed preparation and sowing operations. It is a year round system of managing plant residue with implements that undercut residues, losses the soil and kills the weeds.

Advantages

- Energy conservation through reduced tillage operations.
- Improve the soil physical properties.
- Reduce the water runoff from fields.

**Main field preparation:**

Tillage operations are generally classified in to two, preparatory cultivation and after cultivation. The preparatory cultivation or tillage is operations that are done before the cultivation. This preparatory cultivation is generally called as main field preparation. The main field preparation involves three processes, viz., primary tillage, secondary tillage and lay-out for sowing. Some of the important primary tillage implements are country plough, mould board plough, disc plough, chisel plough etc. Cultivators and harrows are generally used for secondary tillage purpose. However, in practical means, the first two (primary and secondary tillages) may not have any key difference, since; both operations are mainly carried out with same implement. Country plough and cultivators are used for both the purposes. After thorough ploughing, the field modified in to suitable way for planting such as ridges and furrows or beds and channels or pit according to the need of the crops. Such field modifications are mandatory for better crop production.

**Lec. 12. Seeds - Seed rate - Sowing methods - Germination - Crop stand establishment - Planting geometry**

**SEEDS**

Plant propagation is made in two ways, Sexual (by seeds) and asexual (by vegetative means). Biologically, seed is a ripe, fertilized ovule and a unit of reproduction of flowering plants.

**SEED RATE**

The required number of plants/unit area is decided by calculating the seed rate. The seed rate depends on spacing or plant population, test weight, germination percentage. The formula is as follows.

$$\text{Seed rate (kg/ha)} = \frac{\text{Plant population (per ha) x No. of seeds/hill x Test weight (g) x 100}}{\text{1000 x 1000 x Germination percentage (\%)}}$$

**SOWING METHODS**

1. Broadcasting
2. Dibbling
3. Sowing behind the country plough (manual and mechanical drilling)
4. Seed drilling
5. Nursery transplanting

**1. Broadcasting**

Broadcasting is otherwise called as random sowing. Literally means 'scattering the seeds'. Broadcasting is done for many crops. Broadcasting is mostly followed for small sized to medium sized crops. This is the largest method of sowing followed in India, since; it is the easiest and cheapest and requires minimum labours. To have optimum plant population in unit area certain rules should be followed.

- Only a skilled person should broadcast the seeds for uniform scattering.
- The ploughed field should be in a perfect condition to trigger germination.

The seeds are broadcasted in a narrow strip and the sowing is completed strip by strip. To ensure a good and uniform population, it is better to broadcast on either direction. This is called

criss-cross sowing. If the seed is too small, it is mixed with sand to make a bulky one and for easy handling. Ex. Seasame seeds are mixed with sand at 1:15 or 1:10 ratio and sown.

In certain cases the person sowing will be beating the seeds against the basket for uniform scattering. Ex. Sorghum, pearl millet.

After broadcasting, the seeds are covered gently either using a country plough with a very shallow ploughing or some wooden planks (boards / levelers) are used to cover the surface. In some cases, tree twigs or shrub branches are used. If the seeds are large, levelers collect the seeds and leave in the other side. Comb harrow is the best used one.

***Disadvantages***

- All the seeds broadcasted do not have contact with the soil. 100% germination is not possible.
- Enhanced seed rate is required.
- Seeds cannot be placed in desired depth. Desired depth ensures perfect anchorage. Lodging (falling down) is common in broadcasting.

**2. Dibbling**

This is actually line sowing. Inserting a seed through a hole at a desired depth and covering the hole. Dibbling is practiced on plain surface and ridges and furrows or beds and channels. This type of sowing is practiced only under suitable soil condition. Rice fallow cotton is dibbled on a plain surface. The seeds are dibbled at $2/3^{rd}$ from top or $1/3^{rd}$ at bottom of the ridge. Before sowing, furrows are opened and fertilizers are applied above which seeds are sown. The seeds do not have contact with the fertilizers. This is done for wider spaced crops and medium to large sized seeds. Ex. Sorghum, maize, sunflower, cotton are dibbled on ridges and furrows. Both beds and channels; and ridges and furrows come under line sowing. While earthing up, the plant occupies middle of the ridge. Earthing up is essential for proper anchorage of the root system.

Advantages of line sowing are, (i) uniform population, (ii) better germination, (iii) reduced seed rate.

| Sl. No. | Dibbling (Line sowing) | Broadcasting (Random sowing) |
|---|---|---|
| 1. | Costlier | Cheaper |
| 2. | Takes considerable time | Quickest and time saving |
| 3. | Fixed seed rate | Higher seed rate |
| 4. | Mechanization is possible, e.g. weeding, harvesting | Not possible |
| 5. | Uniform utilization of resources (land, water, light, nutrient, etc.) | Resource utilization is un-uniform |

**3. Sowing behind the plough**

Sowing behind the plough is done by manual or mechanical means. Seeds are dropped in the furrows opened by the plough and the same is closed or covered when the next furrow is opened. The seeds are sown at uniform distance. Manual method is a laborious and time consuming process. Seeds like redgram, cowpea and groundnut are sown behind the country plough. Major sown crop is ground out. Seeds are sown by mechanical means by Gorus – seed drill. A seed drill has a plough share and hopper. Seeds are placed on hopper. Different types of seed drill are available, e.g., simple Goru – Guntakas.

***Advantages:*** i) The seeds are placed at desired depth covered by iron planks, ii) except very small, very large seeds most of the seeds can be sown, e.g. maize, sorghum, millets, sunflower, etc.

**4. Drill sowing (or) Drilling**

Drilling is the practice of dropping seeds in a definite depth covered with soil and compacted. In this method, sowing implements are used for placing the seeds into the soil. Both animal drawn Gorus and power operated (seed drills) implements are available. Seeds are drilled continuously or at regular intervals in rows. In this method, depth of sowing can be maintained and fertilizer can also be applied simultaneously. It is possible to take up sowing of inter crops also. It requires more time, energy and cost, but maintains uniform population per unit area. Seeds are placed at uniform depth, covered and compacted.

**5. Transplanting**

This method of planting has two components, a. nursery and b. transplanting. In nursery, young seedlings are protected more effectively in a short period and in a smaller area. Management is easy and economical.

***Advantages***

- Can ensure optimum plant population
- Sowing of main field duration, i.e., management in the main field is reduced
- Crop intensification is possible under transplanting

***Disadvantages***

- Nursery raising is expensive
- Transplanting is another laborious and expensive method

Age of seedlings is $1/4^{th}$ of the total duration of the crop. If the total duration is 16 weeks, four week period (1 month) is under nursery beds. Nursery age is not very rigid, e.g., thumb rule – 3 months crop – nursery duration 3 weeks, minimum 4 months – 4 weeks minimum period; 5 months –

5 weeks. After the nursery period, seedlings are pulled out and transplanted. This is done on the main field after thorough field preparation or optimum tilth. The seedlings are dibbled in lines or in random. Closer spaced crops are mostly raised in random method even after nursery, Ex. rice and finger millet. For vegetables, desired spacing is required during transplanting. Transplanting shock is a period after transplanting, the seedlings show no growth. This is mostly due to the change in the environment between root and the soil. The newly planted seedlings should adjust with new environment. It is for a period of 5-7 days depending upon season, crop, variety, etc. At higher temperature, dehydration is possible and leaves dried out. Area required for nursery normally is $1/10^{th}$ of the total area.

**GERMINATION**

- Germination is a protrusion of radicle or seedling emergence.
- Germination results in the rupture of the seed coat and emergence of seedling from embryonic axis.

**Factors affecting seed germination**

1. Soil: Soil type, texture, structure and microorganisms greatly influence the seed germination.
2. Moisture: When the seeds do not get required moisture in the soil, the viability is lost. When the moisture is excess after germination, it will lead to rotting of the sprouts.
3. Temperature: When it is above and below the optimum temperature, the germination rate will be affected.
4. Light: The most effective wavelength for promoting germination is red (662 nm) and 730 nm inhibits germination.
5. Soil condition: a. Tilth is the most important soil factor influences on germination of seed. Small seeds require fine tilth whereas, moderate and larger seeds requires medium and coarse tilth soils, respectively.

   b. Depth of sowing: The seeds should be placed at optimum depth. When the seeds are placed at deeper layers they have to spend more energy for germination. When it is placed on soil surface, it will be taken away by birds/worked away. The thumb rule is to sow seeds to a depth of approximately 3 to 4 times diameter of the seed. The optimum depth of sowing for most of the field crops ranged between 3 and 5 cm depth. The seeds sown should be protected from rodents or birds before germination by employing labourers to scare the birds at least for three days after sowing.

## CROP STAND ESTABLISHMENT

Good crop establishment is one of the most important features in better crop production. The better crop establishment is in turn expressed as optimum plant population in fields. Number of plants per unit area in the cropped field is called as plant population.

### Optimum plant population

It is the number of plants required to produce maximum output or biomass per unit area. Any increase beyond this stage results in either no increase or reduction in biomass.

### Importance of plant population

- Yield of any crop depends on final plant population.
- The plant population depends on germination percentage and survival rate in the field.
- Under rain fed conditions, high plant population will deplete the soil moisture before maturity, where as low plant population will leave the soil moisture unutilized.
- When soil moisture and nutrients are not limited high plant population is necessary to utilize the other growth factors like solar radiation efficiently.
- Under low plant population, individual plant yield will be more due to wide spacing.
- Under high plant population, individual plant yield will be low due to narrow spacing leading to competition between plants.
- Yield per plant decreases gradually as plant population per unit area is increased, but yield per unit area increases up to certain level of population. That level of plant population is called as optimum population.
- So, to get maximum yield per unit area, optimum plant population is necessary. So the optimum plant population for each crop should be identified.

### Factors affecting plant population

#### A. Genetic Factors

***1. Size of the plant***

- The volume occupied by the plant at the time of flowering decides the spacing of the crop.
- Plants of red gram, cotton, sugarcane etc. occupy larger volume of space in the field compared to rice and wheat.
- Even the varieties of the same crop differ in size of the plant.

***2. Elasticity of the plant***

- Variation in size of the plant between minimum size of the plant that can produce some economic yield to the maximum size of the plant that can reach under unlimited space and resources is the elasticity of the plant.
- The optimum plant population range is high in indeterminate plants. Ex. Redgram – 55,000 to1,33,000 plants/ha.
- The elasticity is due to tillering and branching habit of the plants.
- For determinate plants like pearl millet, sorghum elasticity range is less.
- For indeterminate plants like cotton and Redgram, more branches will be produced by the crop.

***3. Foraging area or soil cover***

- Crop should cover the soil as early as possible so as to intercept maximum sunlight.
- Higher the intercepted radiation more will be the dry matter produced.
- Close spaced crops intercept more solar radiation than wide spaced crops.

***4. Dry matter partitioning***

- Dry matter production is related to amount of solar radiation intercepted by the canopy which depends on plant density.
- As the plant density increases, the canopy expands more rapidly, more radiation is intercepted and more dry matter is produced.

***5. Crop and variety***

Depending on the crops and varieties, the plant population varies.

| | | | |
|---|---|---|---|
| Rice : | Short duration | - | 6,66,666 plants/ha (15 cm x 10 cm) |
| | Medium | - | 5,00,000 plants/ha (20 cm x 10 cm) |
| | Long | - | 3,33,000 plants/ha (20 cm x 15 cm) |
| Cotton : | Medium | - | 55,555 plants/ha (60 cm x 30 cm) |
| | Long | - | 44444 plants/ha (75 cm x 30 cm) |
| | Hybrids | - | 18,518 plants/ha (120 cm x 45 cm) |
| Maize : | Varieties | - | 83,333 plants/ha (60 x 20 cm) |
| | Hybrids | - | 47,620 plants/ha (60 x 35 cm) |

**B. Environmental factors**

***1. Time of sowing***

- The crop is subjected to various weather conditions when sown at different periods.

- Among weather factors, day length and temperature influence much on the plant population. As low temperature retards growth, high plant population is required to cover the soil.

***2. Rainfall / irrigation***

1. Plant population has to be less under rainfed than irrigated condition.
2. Under more plant densities, more water is lost through transpiration.
3. Under adequate rainfall / irrigation, high plant population is recommended.

***3. Fertilizer application***

1. Higher plant population is necessary to fully utilize higher level of nutrients in the soil to realize higher yield.
2. Nutrient uptake increases at optimum plant population.
3. High population under low fertility soils leads to nutrient deficiency symptoms leading low yield.

***4. Seed rate***

- Quantity of seed sown/unit area, viability and establishment rate decides the plant population. Under broadcasting the seed rate is higher when compared with line sowing/transplanting, Ex. Rice.

  Direct sowing - 100 kg/ha; Line sowing - 60 kg/ha; Transplanting - 40 kg/ha.

**PLANT GEOMETRY**

The arrangement of the plants in different rows and columns in an area to utilize the natural resources efficiently is called crop geometry. It is otherwise area occupied by a single plant Ex.. Rice – 20 cm x 15 cm. This is very essential to utilize the resources like light, water, nutrient and space. Different geometries are available for crop production

**Different crop geometries are available for crop production**

***1) Random plant geometry***

Random plant geometry results due to broadcasting method of sowing and no equal space is maintained. Resources are either under utilized or over exploited.

***2) Square plant geometry***

The plants are sown at equal distances on either side. Mostly perennial crops, tree crops follow square method of cultivation. Ex. Coconut – 7.5 x 7.5 m; banana – 1.8 x 1.8 m. But, due to scientific invention, the square geometry concept is expanded to close spaced field crops like rice too.

*Advantages*

Light is uniformly available, movement of wind is not blocked and mechanization can be possible.

***3) Rectangular method of sowing***

There are rows and columns, the row spacing are wider than the spacing between plants.

The different types exist in rectangular method are,

*a) Solid row:* Each row will have no proper spacing between the plants. This is followed only for annual crops which have tillering pattern. There is definite row arrangement but no column arrangement, Ex. Wheat.

*b) Paired row arrangement*: It is also a rectangular arrangement. It a crop requires 60 cm x 30 cm spacing and if paired row is to be adopted the spacing is altered to 90 cm instead of 60 cm in order to accommodate an intercrop. The base population is kept constant.

*c) Skip row*: A row of planting is skipped and hence there is a reduction in population. This reduction is compensated by planting an intercrop; practiced in rainfed or dryland agriculture.

*d) Triangular method of planting:* It is recommended for wide spaced crops like coconut, mango, etc. The number of plants per unit area is more in this system.

**Lec. 13. Weeds – Definition – harmful and beneficial effects of weeds – crop weed competition and management of weeds – IWM**

**WEEDS**

Weeds are the plants, which grow where they are not wanted. Jethro Tul (1731) coined the word weed. Weeds can also be referred to as plants out of place.

***DEFINITION***

Weeds are unwanted and undesirable plant that interfere with utilization of land and water resources and thus, adversely affect crop production and human welfare.

**Characteristics of weeds**

1. In the world about 30,000 species of weeds have been listed.
2. Out of which 18,000 cause serious damage to agricultural production.
3. 18 weeds are considered as world worst weeds.
4. These weed species are more competitive than the cultivated crops.
5. They are capable of thriving under stress condition. They produce enormous seeds.
6. The weeds seeds are easily germinated; they undergo pollination very easily; they have the capacity to withstand adverse situations.

**HARMFUL EFFECTS OF WEEDS**

1. Weeds compete with main crop for space, light, moisture and soil nutrients thus, causing reduction in yield.
2. Affect quality of farm produce, livestock products such as milk and skin.
3. Act as alternate host for pests and pathogens.
4. Cause health problems to human beings. Ex. *Parthenium* causes allergy.
5. Increase cost of cultivation due to weeding problem.
6. Aquatic weeds transpire large quantity of water, obstruct flow of water.
7. Reduce the land value (if *Cynodon* and *Parthenium* are present in the land).
8. Some weeds are poisonous to livestock.

**BENEFICIAL EFFECTS OF WEEDS**

1. Weeds are indirectly responsible for crop cultivation, but for them cultivated crops may not receive much attention

2. Manure: When weeds are ploughed in, they add to the soil plenty of humus. Excellent compost can be made out of many weed plants. e.g. *Calotropis gigantea*, *Croton sparsiflorus* (Syn: *C. bonflandianum*) and *Tephrosia purpurea* are used as green leaf manure for rice.

3. As human food: Weeds serve as human food e.g. *Amaranthus viridis* and *Digera arvensis* used as greens.

4. As fodder: Most weeds are eaten by cattle and weeds like *Rynchosia aurea*, and *Clitoria terneata* are very good fodder legumes.

5. Weed as fuel: *Prosopis juliflora* very invasive in nature and notorious tree weed commonly used as fire wood. People make charcoal out of it and marketed.

6. Weed as soil binders: *Panicum repens* is an excellent soil binder; keeps bunds in position and prevents soil erosion.

7. Weed as medicine: Many weeds have great therapeutic properties and used as medicine. Eg. *Phyllanthus niruri* – Jaundic; *Eclipta alba* – Scorpion sting; *Centella asiatica* – Improves memory; *Cynodon dactylon* – Asthma, piles; *Cyperus rotundus* – Stimulates milk secretion.

8. Weed as mats and screens: Stems of *Cyperus pangorei* and *Cyperus corymbosus* are used for mat making, while *Typha angustata* is used for making screens.

9. Weed as indicators: Weeds are useful as indicators of good and bad soils. *E. colonum* occurs in rich soils while *Cymbopogon* denotes poor light soil and sedges are found in ill-drained soils.

**CROP WEED COMPETITION**

Plants require nutrients, water, light and carbon di-oxide for growth. Light and $CO_2$ are used by the leaves. Competition for these growth factors is, therefore, above the ground. The competition for nutrients and water is below the ground as these are absorbed by roots. At early stage of crop and weeds, leaf area index and weeds density are low and each plant is able to get its requirement as though every one of them is growing in isolation. In majority of crops, the period of non-interference between crop and weeds is very short. Even at low population of weeds, competition is set at three leaf stage of weeds.

As the plants grow, the leaf area index and weeds density increase leading to mutual interference in the absorption of one or more growth factors. Growth rate falls below that obtained from plants gown without interference. The close proximity of weeds and their number cause sub-optimal absorption of growth factors resulting in reduction of growth and yield of crops. Among three factors, light and nutrients are very important in reducing growth.

***Competition for nutrients***

The nutrient concentration of weeds far exceeds the associated crop. Nutrient taken up by maize and its associated weeds show that weeds contain higher concentration of nutrients. About 50% weeds biomass in maize field causes 64-83% of nutrients. In groundnut fields, nutrients

especially P and K are absorbed more (>80% of total nutrients) by weeds. In common experience that crops infested with weeds show nitrogen deficiency.

***Competition for light***

Generally, weeds grow faster and shade the crop plants, if not checked. Even in case of shorter plants, the lower leaves are shaded by weeds. Weeds deplete photosynthetically active radiation resulting in reduction of photosynthesis and shortening the life of lower leaves. Due to shading, crop plants may have thin, pale green leaves with reduced number of vascular bundles and meristem cells.

***Competition for water***

The amount of water required to produce a unit amount of drymatter is known as transpiration ratio. The transpiration ratio of weeds is higher compared to crops. The transpiration ratio of *Cynodon dactylon* is around 813 compared to 450 for pearl millet and 430 for sorghum. This indicates that more water is required to grow one unit of weeds than crops. Weeds consume water luxuriantly and exhaust soil moisture quickly depriving the crop plants of moisture especially under drylands. Weeds consume valuable stored soil moisture from fallow lands. Weeds growing in fallow fields consume as much as 7-12 ha cm of soil moisture which is sufficient to produce 1500-2000 kg of sorghum grains.

***Competition for $CO_2$***

Competition for $CO_2$ may occur when the weed infestation is high. Since most of the weeds are $C_4$ plants, they can deplete very low concentration of $CO_2$ compared to crops.

***Allelopathic effect***

Most of the plants excrete certain chemicals into soil which inhibit germination and growth of other plants in their vicinity and this phenomenon is called as allelopathy. The degree of allelopathic effect depends on the plant species. Important weed species that show allelopathic effect are *Agropyron repens, Sorgum halepense, Lantana camara, Cyperus rotundus* etc.

**Critical period of weed competition**

It is defined as the shortest time span during the crop growth when weeding results in the highest economic returns. The crop yield level obtained by weeding during this period is almost similar to that obtained by full season weed free conditions. The critical period of weed competition is around 30 days for most of the crops.

**Critical period of weed competition for different crops**

| Crop | Critical period (Days after sowing) | |
|---|---|---|
| | From | To |
| Rice (transplanted) | 15 | 45 |
| Upland rice | Entire period | |
| Sorghum | 15 | 45 |
| Maize | 15 | 35 |
| Finger millet | 25 | 45 |
| Soybean | 15 | 45 |
| Blackgram | 30 | 45 |

## METHODS OF WEED CONTROL

### A. Mechanical method

Mechanical or physical methods of weed control are being employed ever since man began to grow crops. The mechanical methods include tillage, hoeing, hand weeding, digging, cheeling, sickling, mowing, burning, flooding, mulching, etc.

#### *1. Tillage*

Tillage removes weeds from the soil resulting in their death. It may weaken plants through injury of root and stem pruning, reducing their competitiveness or regenerative capacity. Tillage also buries weeds. Tillage operation includes ploughing, discing, harrowing and leveling which is used to promote the germination of weeds through soil turnover and exposure of seeds to sunlight, which can be destroyed effectively later. In case of perennials, both top and underground growth is injured and destroyed by tillage.

#### *2. Hoeing*

Hoe has been the most appropriate and widely used weeding tool for centuries. It is a very useful implement to obtain results effectively and cheaply. It supplements the cultivator in row crops. Hoeing is particularly more effective on annuals and biennials as weed growth can be completely destroyed. In case of perennials, it destroyed the top growth with little effect on underground plant parts resulting in re-growth.

#### *3. Hand weeding*

It is done by physical removal or pulling out of weeds by hand or removal by implements called Khurpi, which resembles sickle. It is probably the oldest method of controlling weeds and it is

still a practical and efficient method of eliminating weeds in cropped and non-cropped lands. It is very effective against annuals, biennials and controls only upper portions of perennials.

***4. Digging***

Digging is very useful in the case of perennial weeds to remove the underground propagating parts of weeds from the deeper layer of the soil.

***5. Cheeling***

It is done by hand using a cheel hoe, similar to a spade with a long handle. It cuts and shapes the above ground weed growth.

***6. Sickling and mowing***

Sickling is also done by hand with the help of sickle to remove the top growth of weeds to prevent seed production and to starve the underground parts. It is popular in sloppy areas where only the tall weed growth is sickled leaving the root system to hold the soil in place to prevent soil erosion.Mowing is a machine-operated practice mostly done on roadsides and in lawns.

***7. Burning***

Burning or fire is often an economical and practical means of controlling weeds. It is used to (a) dispose of vegetation, (b) destroy dry tops of weeds that have matured, (c) kill green weed growth in situations where cultivations and other common methods are impracticable.

***8. Flooding***

Flooding is successful against weed species sensitive to longer periods of submergence in water. Flooding kills plants by reducing oxygen availability for plant growth. The success of flooding depends upon complete submergence of weeds for longer periods.

***Merits of mechanical method***

a. Oldest, effective and economical method, b. Large area can be covered in shorter time, c. Safe method for environment, d. Does not involve any skill, e. Weeding is possible in between plants, f. Deep rooted weeds can be controlled effectively

***Demerits mechanical method***

a. Labour consuming, b. Possibility of damaging crop, c. Requires ideal and optimum specific condition.

**B. Cultural weed control**

Several cultural practices like tillage, planting, fertiliser application, irrigation etc., are employed for creating favourable condition for the crop. These practices if used properly, help in controlling weeds. Cultural methods, alone cannot control weeds, but help in reducing weed population. They should, therefore, be used in combination with other methods. In cultural methods,

tillage, fertiliser application and irrigation are important. In addition, aspects like selection of variety, time of sowing, cropping system, cleanliness of the farm etc., are also useful in controlling weeds.

***1. Field preparation***

The field has to be kept weed free. Flowering of weeds should not be allowed. This helps in prevention of build up of weed seed population.

***2. Summer tillage***

The practice of summer tillage or off-season tillage is one of the effective cultural methods to check the growth of perennial weed population in crop cultivation. Initial tillage before cropping should encourage clod formation. These clods, which have the weed propagules, upon drying desiccate the same. Subsequent tillage operations should break the clods into small units to further expose the shriveled weeds to the hot sun.

***3. Mulching***

Mulch is a protective covering of material maintained on soil surface. Mulching has smothering effect on weed control by excluding light from the photosynthetic portions of a plant and thus inhibiting the top growth. It is very effective against annual weeds and some perennial weeds like *Cynodon dactylon.* Mulching is done by dry or green crop residues, plastic sheet or polythene film. To be effective the mulch should be thick enough to prevent light transmission and eliminate photosynthesis

***4. Solarisation***

This is another method of utilisation of solar energy for the desiccation of weeds. In this method, the soil temperature is further raised by 5-10 $^{o}$C by covering a pre-soaked fallow field with thin transparent plastic sheet. The plastic sheet checks the long wave back radiation from the soil and prevents loss of energy by hindering moisture evaporation.

***5. Stale seedbed***

A stale seedbed is one where initial one or two flushes of weeds are destroyed before planting of a crop. This is achieved by soaking a well prepared field with either irrigation or rain and allowing the weeds to germinate. At this stage a shallow tillage or non- residual herbicide like paraquat may be used to destroy the dense flush of young weed seedlings. This may be followed immediately by sowing. This technique allows the crop to germinate in almost weed-free environment.

***6. Blind tillage***

The tillage of the soil after sowing a crop before the crop plants emerge is known as blind tillage. It is extensively employed to minimise weed intensity in drill sowing crops where emergence

of crop seedling is hindered by soil crust formed on receipt of rain or irrigation immediately after sowing

***Merits of Cultural Method***

a. Low cost for weed control, b. Easy to adopt, c. No residual problem, d. Technical skill is not involved, e. No damage to crops, f. Effective weed control, g. Crop-weed ecosystem is maintained.

***Demerits of Cultural Method***

a. Immediate and quick weed control is not possible, b. Weeds are kept under suppressed condition, c. Perennial and problematic weeds can not be controlled, d. Practical difficulty in adoption.

**C. Biological weed control**

Use of living organisms viz., insects, disease organisms, herbivorous fish, snails or even competitive plants for the control of weeds is called biological control. In biological control method, it is not possible to eradicate weeds but weed population can be reduced. This method is not useful to control all types of weeds. Introduced weeds are best targets for biological control.

Examples of biological control: Water haycinth is controlled by Haycinth moth; *Opuntia* is controlled by scale insects; *Parthenium* is by *Zygogramma* beetle; Strangler vine (*Morrenia odorata*) controlled by fungus- *Phytophthora palmivora* (Devine) in citrus gardens Devine is a bio-herbicide

**D. Chemical weed control**

Using chemicals, generally referred as herbicides, for the control of weeds is called chemical weed control. In 1944, discovery of 2,4-D Na salt as a land mark in herbicide usage. Commonly used herbicides are Pendimethalin, Fluchloralin, Atrazine.

***Merits of chemical weed control***

- Herbicide can be recommended for adverse soil and climatic conditions, as manual weeding is highly impossible during monsoon season.
- Herbicide can control weeds even before they emerge from the soil so that crops can germinate and grow in completely weed-free environment at early stages. It is usually not possible with physical weed comfort.
- Weeds, which resemble like crop in vegetative phase, may escape in manual weeding. However, these weeds are controlled by herbicides.
- Herbicide is highly suitable for broadcasted and closely spaced crops.

- Controls the weeds without any injury to the root system of the associated standing crop especially in plantation crops like Tea and Coffee
- Reduces the need for pre planting tillage
- Controls many perennial weed species
- Herbicides control the weed in the field itself or *in-situ* controlling whereas mechanical method may lead to dispersal of weed species through seed
- It is profitable where labour is scarce and expensive
- Suited for minimum tillage concept and highly economical

***Demerits***

- Pollutes the environment
- Affects the soil microbes if the dose exceeds
- Herbicide causes drift effect to the adjoining field
- It requires certain amount of minimum technical knowledge for calibration
- Leaves residual effects
- Some herbicide is highly costlier
- Suitable herbicides are not available for mixed and inter-cropping system.

**Integrated weed management**

**FAO Definition**

It is a method whereby all economically, ecologically and toxicologically justifiable methods are employed to keep the harmful organisms below the threshold level of economic damage, keeping in the foreground the conscious employment of natural limiting factors.

- Uses a variety of technologies in a single weed management with the objective to produce optimum crop yield at a minimum cost taking into consideration ecological and socio-economic constraints under a given agro-ecosystem.

Weeds can be controlled by several methods. However, each weed control method has advantages and disadvantages. None is applicable in all crops and weed situations. Continuous use of same method leads to build up of tolerant weeds. It is, therefore, necessary to combine or change the method of weed control. Integrated weed management is a weed population management system that uses all suitable techniques in a combative manner to reduce weed population and maintain them at levels below those causing economic injury. The components of integrated weed management are cultural, physical, biological and chemical methods. By using two or more of these methods, suitable for crop and weed situation, weeds can be controlled effectively and economically.

**Lec. 14. Role of manures and fertilizers in crop production – agronomic interventions for enhancing FUE - Inter cultivation - Thinning - Gap filling and other intercultural operations**

**MANURES**

Manures are plant and animal wastes that are used as source of plant nutrients. They release nutrients after their decomposition. Manures can be grouped into bulky organic manures and concentrated organic manures.

**FERTILIZERS**

Fertilizers are industrially manufactured chemical containing plant nutrients. Nutrient content is higher in fertilizers than organic manures and nutrients are released almost immediately.

**ROLE OF MANURES AND FERTILIZERS**

1. Organic manures bind the sandy soil and improve its water holding capacity.
2. Organic manures open the clayey soil and help in aeration better root growth.
3. Organic manures add plant nutrients in small percentage and also add micro nutrients which are essential for plant growth. The microbial activity is increased which helps in releasing plant nutrients in available. Ex. Bulky organic manures- FYM, compost from organic waste, night soil, sludge, sewage, sheep folding, green manures. Concentrated organic manures - oilcakes (edible, non-edible), blood meal, fish meal and bone meal.
4. Organic manures should be incorporated before the sowing or planting because of slow release of nutrients.
5. Fertilizers play an important role in crop production as the nutrient elements in fertilizer are present in higher concentration and can be readily utilize by plants directly or after rapid transformation.
6. Fertilizers dose can be adjusted to suit the requirement as determine by soil fertility evaluation. Fertilizers applied through straight fertilizers (providing single nutrient) or complex and mixed fertilizers (supplies two or more nutrients).

**AGRONOMIC INTERVENTIONS FOR ENHANCING FUE**

The following are the agronomic measures to improve the Fertilizer use efficiency (FUE).

1. Using best fertilizer source
2. Using adequate rate & diagnostic techniques
3. Usage of balanced fertilization
4. Integrated nutrient management
5. Utilization of residual nutrients

**1. Using best fertilizer source:**

Identification of best source of fertilizer is pre-requisite for better crop production. Source of fertilizer depends on crop and variety, climatic and soil condition, availability of fertilizer, etc.

- Nitrogen: Ammoniacal or Nitrate
- Phosphorus: Water soluble or Citrate soluble
- Potassium: Muriate of potash
- Sulphur: Sulphate or Elemental S
- Multinutrient fertilizers: MAP, DAP, SSP, Nitrophosphates
- Multi-nutrient mixtures: Several combinations of NPK
- Fortified fertilizers: Neem-coated urea, Zincated urea, Boronated SSP, NPKS mix.

**2. Using adequate rate & diagnostic techniques:**

The fertilizer recommendation must be adequate quantity so as to meet the demand of crop at any point of growth. The fertilizer supply is made by diagnosing its requirement by any of the following method.

- State recommended generalized fertilizer dose or blanket recommendation
- Soil-test based fertilizer recommendations
- Soil-test crop response based recommendation
- Plant analysis for diagnosing nutrient deficiencies
- Chlorophyll meter and Leaf colour charts, etc.

**3. Balanced fertilization**

Balanced fertilization includes adequate supply of all essential nutrients, proper method of application, right time of application and nutrient interrelationships.

**a. Adequate supply of all essential nutrients:** Due to more concentration and application on primary nutrients (NPK), soils developed deficiency symptoms for secondary and micro-nutrients. Hence, ignored elements must be added with the NPK (may be in minor quantity) to get higher yields in crops. Experimental results shown that about 20-25 kg of micro-nutrient application or two foliar sprays increases the yield of crops up to 20%.

**b. Proper method:** N and K can be applied as broadcasting and band placement. Water soluble P fertilizers are preferred to apply as band placement in neutral & alkaline soils. Citrate soluble P fertilizers are applied as broadcast method in acidic soils. Sulphate forms of S fertilizers are applied as broadcasting or band placement, whereas, elemental S and pyrite are applied as broadcasting method. Micronutrients are applied in minor quantity as foliar sprays and water soluble fertilizers are applied in fertigation.

**c. Right time:**

- Upland crops - 2 splits (seeding, 3-5 weeks after first dose)
- Flooded rice - 3 splits (Transplanting, 3 and 6 weeks after first dose)

**d. Nutrient interrelationships:**

**A**ntagonistic nature of fertilizers is to be considered while applying into the soil. Some of the fertilizer application in excess causes reduction of yield and quality of crops. Ex. Application of excessive 120 kg P $ha^{-1}$ created an imbalance and reduced the seed and oil yields in soybean compared to 80 kg P $ha^{-1}$.

**4. Integrated nutrient management**

Organic manures, crop residues, green manures, bio-fertilizers etc. are to be blended in right manner along with inorganic fertilizers to meet the crop demand. All the possible and available organic sources are to be utilized efficiently to reduce the usage of inorganic fertilizers.

**5. Utilization of residual nutrients**

Some of the strategies to utilize the crop residues in efficient manner are,

- Knowledge on climatic conditions & carry-over effects of residues.
- Blending rightly on cereal-legume rotations
- Mixing shallow-deep rooted crop rotations

**INTER CULTIVATION**

Cultivation practices taken up after sowing of crop is called inter-cultivation. It is otherwise called as after operation. There are three important after cultivation processes viz., Thinning and gap filling, weeding and hoeing and earthing up.

**1. Thinning and Gap filling**

The objective of thinning and gap filling process is to maintain optimum plant population. Thinning is the removal of excess plants leaving healthy seedlings. Gap filling is done to fill the gaps by sowing of seeds or transplanting of seedlings in gap where early sown seed had not germinated. It is a simultaneous process. Normally, these are practiced a week after sowing to a maximum of 15 days. In dryland agriculture, gap filling is done first. Seeds are dibbled after 7 days of sowing. Thinning is done after gap filling in order to avoid drought. It is a management strategy to remove a portion of plant population to mitigate stress is referred to as mid season correction.

**2. Weeding and Hoeing**

Weeding is removal of unwanted plants. Weeding and hoeing is a simultaneous operation. Hoeing is disturbing the top soil by small hand tools and helps in aerating the soil.

**3. Earthing up**

It is a dislocation of soil from one side of a ridge and to be placed nearer the cropped side. It is carried out in wide spaced and deep rooted crops. It is done around 6-8 weeks after sowing / planting in sugarcane, tapioca, banana, etc.

**4. Other inter cultivation practices**

**Harrowing**

Stirring or scraping the surface soil in inter and intra row spacing of the crop using tools or implements.

**Roguing**

Removal of plants of a variety admixed with other variety of same crop. Ex. In IR 50 rice field, the other rice varieties are rogue. It is practiced in seed production to maintain purity.

**Topping**

Removal of terminal buds. It is done to stimulate auxillary growth. Practiced in cotton and tobacco.

**Propping**

Provision of support to the crop is called propping. Practiced in sugarcane commonly. It is done to prevent lodging of the crop. Cane stalks from adjacent rows are brought together and tied with their own trash and old leaves.

**De-trashing**

Removing of older leaves from the sugarcane crop.

**De-suckering**

Removal of axillary buds and branches which are considered non essential for crop production and which removes plant nutrients considerably are called suckers. Ex. Tobacco.

**Lec. 15. Irrigation - Time and methods - Modern techniques of irrigation - Drainage and its importance**

**IRRIGATION**

Irrigation is defined as the artificial application of water to the soil for the purpose of crop production in supplement to rainfall and ground water contribution.

**Importance of water to plants**

1. Plants contain 90% water which gives turgidity and keeps them erect
2. Water is an essential part of protoplasm
3. It regulates the temperature of the plant system
4. It is essential to meet the transpiration requirements
5. It serves as a medium for dissolving the nutrients present in the soil
6. It is an important ingredient in photosynthesis

**TIME OF IRRIGATION**

Crops draw water from the moisture stored in the soil. When the moisture present in the soil is low, then the plant requirements are not met with. When the soil is supplied with moisture in excess the supply of air is reduced that limits the plants growth. In between, there is a range of moisture content that is called as optimum soil moisture range for plant growth.

The upper limit of the optimum soil moisture range is the field capacity (-0.01 to -0.03 Mpa) and the lower limit is just above the wilting point (-1.5 Mpa). The objective of irrigation is to store the moisture in the soil between these limits. Immediately after irrigation, all the macro and micropores will be filled with water. All the water present in macropores will drain to subsoil within 48 hours and moisture in micropores will be available to plants.

As the soil dries due to loss of water by evaporation and transpiration, the plant wilts during day time to conserve moisture and become normal at night. When the same condition continues, the plant will wilt without dieing. This condition is called as wilting coefficient. The irrigation is given whenever plants require water. This is decided by the crop and soil appearance.

**IRRIGATION METHODS**

1. Surface 2. Sub-surface 3. Pressurized irrigation

Criteria for selection of irrigation method

- Water supply source
- Topography
- Quantity of water to be applied

- The crop
- Method of cultivation

**Surface irrigation methods**

Oldest (4000 years back) and most common method. 90% of world's total irrigated area is under this method. In USA also, 66% is by surface irrigation. This method is most suitable for low to moderate infiltration rates and leveled lands and having <2-3% slope. It is labour intensive.

Surface is grouped as Border, check basin and furrow irrigations. Border is again classified in to two as straight and contour. Check basins may be of rectangular, contour or ring, furrow irrigation is classified as deep furrow and corrugated furrows. These may be again straight or contour according to direction and leveled and graded as per their elevation.

**1. Border irrigation**

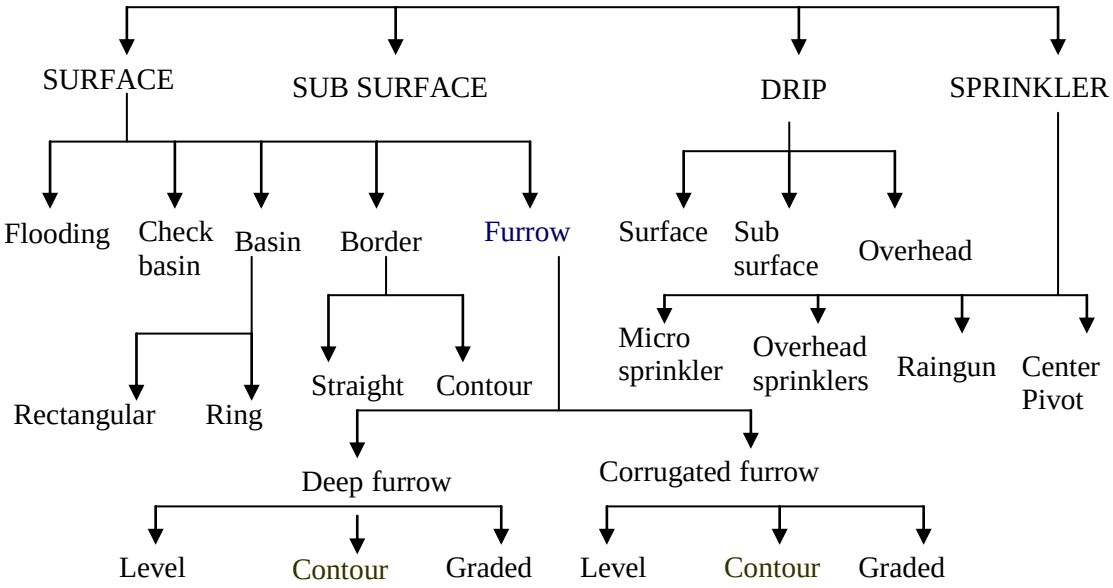
The land is divided into number of long parallel strips called borders. These borders are separated by low ridges. The border strip has a uniform gentle slope in the direction of irrigation. Each strip is irrigated independently by turning the water in the upper end. The water spreads and flows down the strip in a sheet confined by the border ridges.

***Suitability*:** Suitable to soils having moderately low to moderately high infiltration rates. It is not used in coarse sandy soils that have very high infiltration rates and also in heavy soils having very low infiltration rate. Suitable to irrigate all close growing crops like wheat, barley, fodder crops and legumes and not suitable for rice

***Advantages***

a) Border ridges can be constructed with simple farm implements like bullock drawn "A" frame ridger or bund former.
b) Labour requirement in irrigation is reduced as compared to conventional check basin method.
c) Uniform distribution of water and high water application efficiencies are possible.
d) Large irrigation streams can be efficiently used.
e) Adequate surface drainage is provided if outlets are available.

IRRIGATION METHODS




**2. Check basin irrigation**

It is the most common surface irrigation method. Here, the field is divided into smaller unit areas so that each has a nearly level surface. Bunds or ridges are constructed around the area forming basins within which the irrigation water can be controlled. The water applied to a desired depth can be retained until it infiltrates into the soil. The size of the basin varies from 10 $m^2$ to 25 $m^2$ depending upon soil type, topography, stream size and crop.

***Adaptability***

- Small gentle and uniform land slopes and soils having moderate to low infiltration rates.
- Adapted to grain and fodder crops in heavy soils and suitable to permeable soils

***Advantages***

- Check basins are useful when leaching is required to remove salts from the soil profile.
- Rainfall can be conserved and soil erosion is reduced by retaining large part of rain
- High water application and distribution efficiency.

***Limitations***

- The ridges interfere with the movement of implements.
- More area occupied by ridges and field channels.
- The method impedes surface drainage

- Precise land grading and shaping are required
- Labour requirement is higher.
- Not suitable for crops which are sensitive to wet soil conditions around the stem

**3. Furrow irrigation**

It is used for row crops. The furrows are formed between crop rows. The dimension of furrows depend on the crop grown, equipment used and soil type. Water is applied by small running streams in furrows between the crop rows. Water infiltrates into soil and spreads laterally to wet the area between the furrows. In heavy soils, furrows can be used to dispose the excess water.

***Adaptability***

- Method used for wide spaced row crops including vegetables.
- Suitable for maize, sorghum, sugarcane, cotton, tobacco, groundnut, potatoes
- Suitable to most soils except sandy.

***Advantages***

- Water in furrows contacts only 1/2 to 1/5 of the land surface.
- Labour requirement for land preparation and irrigation is reduced.
- Compared to check basins, there is less wastage of land in field ditches

There are three types of furrow irrigation, they are, all furrow irrigation, alternate furrow irrigation and skip furrow irrigation

**4. Surge irrigation**

- Surge irrigation is the application of water into the furrows intermittently in a series of relatively short ON and OFF times of irrigation cycle.
- It has been found that intermittent application of water reduces the infiltration rate over surges thereby the water front advances quickly. Hence, reduced net irrigation water requirement.
- This also results in more uniform soil moisture distribution and storage in the crop root zone compared to continuous flow.
- The irrigation efficiency is in between 85 and 90%.

**II. SUB-SURFACE IRRIGATION**

In subsurface irrigation, water is applied beneath the ground by creating and maintaining an artificial water table at some depth, usually 30-75 cm below the ground surface. Moisture moves upwards towards the land surface through capillary action. Water is applied through underground

field trenches laid 15-30 m apart. Open ditches are preferred because they are relatively cheaper and suitable to all types of soil. The irrigation water should be of good quality to prevent soil salinity.

***Advantages***

- Minimum water requirement for raising crops
- Minimum evaporation and deep percolation losses
- No wastage of land
- No interference to movement of farm machinery
- Cultivation operations can be carried out without concern for the irrigation period.

***Disadvantages***

- Requires a special combination of natural conditions.
- There is danger of water logging
- Possibility of choking of the pipes lay underground.
- High cost.

**III. PRESSURIZED OR MODERN IRRIGATION SYSTEMS**

**a. Drip irrigation system**

Or trickle irrigation is one of the latest and modern methods of irrigation. It is suitable for water scarcity and salt affected soils. Water is applied in the root zone of the crop. Standard water quality test needed for design and operation of drip irrigation system.

***Drip components***

- A drip irrigation system consists of a pump or overhead tank, main line, sub-mains, laterals and emitters.
- The mainline delivers water to the sub-mains and the sub-mains into the laterals.
- The emitters which are attached to the laterals distribute water for irrigation.
- The mains, sub-mains are usually made of PVC (poly vinyl chloride) pipes and and laterals of LLDPE tubes. The emitters are also made of PVC material.
- The other components include pressure regulator, filters, valves, water meter, fertilizer application devices etc.

***Advantages of drip irrigation***

- High water use efficiency (~95%, compared to less than 50% in surface)
- Flexibility of wetted area
- Versatile selection of emitters: type, discharge rate, position

- Economy in weed control
- Low interference with cultivation
- Day and night irrigation
- Prevention of leaf wetting
- Energy saving
- Salinity control
- Irrigation at variable topographic conditions.

***Limitation of drip Irrigation***

- High investment
- High level of knowledge for optimal and economical operation
- Susceptibility to mechanical damage
- Large number of emitters
- Long application time
- High level of filtration and other controls
- Maintenance.

**b. Sprinkler irrigation system**

This is another important modern irrigation techniques followed all over the globe. Sprinkler irrigation is application simulating rainfall overhead so overhead sprinklers. The sprinkler (overhead or pressure) irrigation system conveys water to the field through pipes (aluminium or PVC) under pressure with a system of nozzles. This system is designed to distribute the required depth of water uniformly, which is not possible in surface irrigation. Water is applied at a rate less than the infiltration rate of the soil hence the runoff from irrigation is avoided.

***Types of sprinkler system***

On the basis of arrangement for spraying irrigation water, they are classified as, rotating head (or) revolving sprinkler system and perforated pipe system

Based on the portability, sprinkler systems are classified as, portable system, semi permanent system, solid set system and permanent system.

***Advantages of sprinkler***

- Suitable for undulating topography (sloppy lands)
- Water saving (35-40%) compared to surface irrigation methods.
- Fertilizers and other chemicals can be applied through irrigation water
- Saving in fertilizers, even distribution and avoids wastage.

- Reduces erosion
- Suitable for coarse textured soils (sandy soils)
- Frost control - protect crops against frost and high temperature
- Drainage problems eliminated
- Saving in land

***Limitations***

- High initial cost
- Efficiency is affected by wind
- Higher evaporation losses in spraying water
- Not suitable for tall crops like sugarcane
- Not suitable for heavy clay soils
- Poor quality water can not be used (Sensitivity of crop to saline water and clogging of nozzles)

**Drainage**

Drainage is the artificial removal of water in excess of the quantity required for the crop. Drainage includes removal of excess water of both surface and subsurface in the root zone of crops. Irrigation and drainage go together and are not mutually exclusive. Irrigation aims at supplying optimum quantities of water throughout the crop period, whereas, drainage aims at removing excess quantity of water in a short time. Often, both may be required together to assure sustained and high level production of crops.

**Role of drainage**

- Draining the land provides conditions favorable for crop production.
- The greatest benefit of drainage relates to aeration. Good drainage facilitates the ready diffusion of oxygen to the root zone and escape of carbon dioxide from the root zone into the atmosphere. Several harmful gases also escape from the root zone into the atmosphere.
- The activity of aerobic organisms which influence the availability of nutrients such as nitrogen and sulphur to plants depends on soil aeration and hence, drainage improves aerobic organisms.
- Toxicity in acid soils due to excess iron and manganese is decreased by drainage (due to presence of oxygen in the root zone).
- Drainage permits roots to grow deeper and spread wider thereby increasing the volume of soil from which nutrients can be extracted.

- The removal of excess water helps in drying of the soil quickly and optimum soil temperature permits timeliness of field operations.
- The provision of a good drainage system permits the removal of excess salts in the soil or irrigation water and prevents their build up in the upper soil layers.

**Lec. 16. Cropping pattern and cropping system - Intensive cropping- Sustainable agriculture – IFS**

## CROPPING PATTERN AND CROPPING SYSTEM

Traditionally, increased food production has come from putting more land under cultivation. However, in large areas of the world, especially in Asia, all the land that can be economically cultivated is already in use. In future, most of the extra food needs must come from higher production from land already being farmed. A major share of this increase is likely to come from increasing the number of crops produced per year on a given land using improved crop cultivars. Such multiple cropping offers potential not only to increase food production but also land degradation.

In India, the concept of cropping systems is as old as agriculture. Farmers preferred mixed cropping, especially under dry land conditions, to minimise the risk of total crop failure. Even in Vedas, there is a mention of first and second crops, indicating the existence of sequential cropping.

A **system** is defined as a set of components that are interrelated and interact among themselves. A **cropping system** refers to a set of crop systems, making up the cropping activities of a farm system. Cropping system comprises all components required for the production of a particular crop and the interrelationships between them and environment (TAC, CGIAR, 1978). In other words, a cropping system usually refers to a combination of crops in time and space. Combination in time occurs when crops occupy different growing period and combinations in space occur when crops are inter planted. When annual crops are considered, a cropping system usually means the combination of crops within a given year (Willey *et al.,* 1989

### Cropping pattern

The yearly sequence and spatial arrangement of crops or of crops and fallow on a given area.

### Cropping system

The cropping patterns used on a farm and their interaction with farm resources, other farm enterprises, and available technology which determine their make up.

## INTENSIVE CROPPING

### Principles

The turn around period between one crop and another is minimised through modified land preparation. It is possible when the resources are available in plenty. Ex. Garden land cultivation. Cropping intensity is higher in intensive cropping system. Crop intensification technique includes

intercropping, relay cropping, sequential cropping, ratoon cropping, etc. All such systems come under the general term multiple cropping.

**Need for intensive cropping**

- Cropping systems has to be evolved based on climate, soil and water availability for efficient use of available natural resources.
- The increase in population has put pressure on land to increase productivity per unit area, unit time and for unit resource used.
- This cropping system should provide enough food for the family, fodder for cattle and generate sufficient cash income for domestic and cultivation expenses.

**Intensive cropping:** Growing number of crops on the same piece of land during the given period of time.

**Cropping intensity:** Number of crops cultivated in a piece of land per annum is cropping intensity. In Punjab and Tamil Nadu, the cropping intensity is more than 100% (i.e. around 140-150%). In Rajasthan, the cropping intensity is less.

**Multiple cropping:** The intensification of cropping in time and space dimensions. Growing two or more crops on the same field in a year.

**Forms of multiple cropping**

**Intercropping:** Growing two or more crops simultaneously on the same field. Crop intensification is in both time and space dimensions. There is intercrop competition during all or part of crop growth.

(a) *Mixed intercropping:* Growing two or more crops simultaneously with no distinct row arrangement. Also referred to as mixed cropping. Ex: Sorghum, pearl millet and cowpea are mixed and broadcasted in rainfed conditions.

(b) *Row intercropping:* Growing two or more crops simultaneously where one or more crops are planted in rows. Often simply referred to as intercropping. Maize + greengram (1:1), Maize + blackgram (1:1), Groundnut + Rredgram (6:1)

(c) *Strip intercropping:* Growing two or more crops simultaneously in strips wide enough to permit independent cultivation but narrow enough for the crops to interact agronomically. Ex. Groundnut + redgram (6:4) strip.

(d) *Relay intercropping:* Growing two or more crops simultaneously during the part of the life cycle of each. A second crop is planted after the first crop has reached its reproductive stage of growth, but, before it is ready for harvest. Often simply referred to as relay cropping. Rice- rice fallow pulse.

**Advantages of intercropping**

- Better use of growth resources including light, nutrients and water
- Suppression of weeds
- Yield stability; even if one crop fails due to unforeseen situations, another crop will yield and gives income
- Successful intercropping gives higher equivalent yields (yield of base crop + yield of intercrop), higher cropping intensity
- Reduced pest and disease incidences
- Improvement of soil health and agro-eco system

**Sequential cropping:** Growing two or more crops in sequence on the same field in a farming year. The succeeding crop is planted after the preceding crop has been harvested. Crop intensification is only in time dimension. There is no intercrop competition.

(a) *Double, triple and quadruple cropping:* Growing two, three and four crops, respectively, on the same land in a year in sequence.

Ex. Double cropping: Rice: cotton; Triple cropping: Rice: rice: pulses; Quadruple cropping: Tomato: ridge gourd: *Amaranthus* greens: baby corn

(b) *Ratoon cropping:* The cultivation of crop re-growth after harvest, although not necessarily for grain. Ex. Sugarcane: ratoon; Sorghum: ratoon (for fodder).

The various terms defined above bring out essentially two underlying principles, that of growing crops simultaneously in mixture, i.e., intercropping; and of growing individual crops in sequence, i.e., sequential cropping. The cropping system for a region or farm may comprise either or both of these two principles.

**SUSTAINABLE AGRICULTURE**

Definition:

A farming systems that are "capable of maintaining their productivity and usefulness to society indefinitely and must be resource-conserving, socially supportive, commercially competitive, and environmentally sound."

**USDA (legal)**

Sustainable agriculture means, an integrated system of plant and animal production practices having a site-specific application that will, over the long term:

- satisfy human food and fiber needs;
- enhance environmental quality and the natural resource based upon which the agricultural economy depends;

- make the most efficient use of nonrenewable resources and on-farm resources and integrate, where appropriate, natural biological cycles and controls;
- sustain the economic viability of farm operations;
- enhance the quality of life for farmers and society as a whole.

**Advantages**

- Production cost is low
- Over all risk of the farmer is reduced
- Pollution of water is avoided
- Very little or no pesticide residue is ensured
- Ensures both short and long term profitability

**Disadvantages**

- Since sustainable agriculture uses least quantum of inputs, naturally the output (yield) may also be less.

**Major components of sustainable agricultural system**

- Soil and water conservation to prevent degradation of soil productivity
- Efficient use of limited irrigation water without leading to problems of soil salinity, alkalinity and high ground water table
- Crop rotations that mitigate weed, disease and insect problems, increase soil productivity and minimise soil erosion
- Integrated nutrient management that reduces the need for chemical fertilizers improves the soil health and minimise environmental pollution by conjunctive use of organics, in-organics and bio-fertilizers.
- 'Integrated pest management that reduces the need for agrochemicals by crop rotation, weather monitoring, use of resistant cultivar, planting time and biological pest control.
- Management system to control weed by preventive measures, tillage, timely inter cultivation and crop rotation to improve plant health.

**INTEGRATED FARMING SYSTEM (IFS)**

Integration of two or more appropriate combination of enterprises like crop, dairy, piggery, fishery, poultry, bee keeping etc., for each farm according to the availability of resources to sustain and satisfy the necessities of the farmer

**Definition:** A farming system is a collection of distinct functional units such as crop, livestock, processing, investments and marketing activities which interact because of the joint use of inputs they receive from the environment which have the common objective of satisfying the farmers' (decision makers) aims. The definition of the borders of the options depends on circumstances; often it includes not only the farm (economic enterprise) but also the household (farm – household system)"

**Possible enterprises**

Wetland based farming system

- Crop + Fish + Poultry/poultry/pigeon
- Crop + Fish + Mushroom

**Gardenland based farming system**

- Crop + Dairy + Biogas
- Crop + Dairy + Biogas + Sericulture
- Crop + Dairy + Biogas + Mushroom + Sylvi-culture

**Dry land based farming system**

- Crop + Goat + Agroforestry
- Crop + Goat + Agroforestry + Horticulture

**Benefits of IFS**

- Higher Productivity
- Profitability
- Sustainability
- Balanced food
- Recycling reduces pollution
- Money round the year
- Employment generation
- Increase input efficiency
- Standard of living of the farmer increased
- Better utilisation of land, labour, time and resources

## Lec. 17. Organic / eco-friendly agriculture - Dry farming - Concepts and principles

**Organic farming:** Organic farming is a production system where all kinds of agricultural products are produced organically, including grains, meat, dairy, eggs, fibers such as cotton, flowers and processed food products.

Organic farming avoids or largely excludes the use of synthetic fertilizers, pesticides, growth regulators and livestock feed additives.

**Components of organic farming**

It largely rely upon crop rotations, crop residues, animal manure, legumes, green manure, on/off farm organic wastes, mechanical cultivation, mineral bearing rocks and aspects of biological control of pests and diseases to maintain soil productivity and tilth to supply plant nutrients.

**Scope**

1. It is a sustainable agricultural system which maintains and improves soil fertility so as to guarantee for adequate food security in the future.
2. It relies upon resources from its own area which is not depends much on imported resources.
3. It guides the working of biological process in natural ecosystem.

**Concepts**

1. Building up of biological soil fertility
2. Control of pests, diseases and weeds through development of an ecological balance within the system and by use of bioagents and various cultural techniques.
3. It recycles all wastes and manure within the farm.

**Components:** The components of organic farming are,

- Organic manures
- Non- chemical weed control measures
- Biological pest and disease management.

**How do organic farmers fertilize crops?**

- Organic farmers build healthy soils by nourishing the microbial inhabitants that release, transform and transfer nutrients.
- Soil organic matter contributes to good soil structure and water-holding capacity.
- Organic farmers feed soil biota and build soil structure and enhance water-holding capacity.
- Organic farmers build soil organic matter with cover crops, compost and biologically based soil amendments.

**How do they control pests, diseases and weeds?**

- They produce healthy plants that are better able to resist disease and insect predation.
- Organic farmers' primary strategy in controlling pests and diseases is prevention through good plant nutrition and management.
- Organic farmers use cover crops and sophisticated crop rotations to change the field ecology, effectively disrupting habitat for weeds, insects and disease organisms.
- Weeds are controlled through crop rotation, mechanical tillage, and hand-weeding, as well as through cover crops, mulches and other management methods.
- Organic farmers rely on a diverse population of soil organisms, beneficial insects and birds to keep pests in check.
- Growers implement a variety of strategies such as the use of insect predators, mating disruption, traps and barriers.

**Synonyms of organic farming:**

Eco-farming, biological farming, bio-dynamic farming and macrobiotic agriculture are some of the synonyms of organic farming.

**Eco-farming:** Farming in relation to ecosystem. It is the potential for introducing mutually reinforcing ecological approaches to food production. It aims at the maintenance of soil chemically, biologically and physically the way nature would do it left alone. Soil would then take proper care of plants growing on it. ***Feed the soil, not the plant*** is the watchword and slogan of ecological farming.

**Biological farming:** Farming in relation to biological diversity.

**Biodynamic farming:** Farming which is biologically organic and ecologically sound and sustainable farming.

Organic farming is not exclusively based on short term economics but also considers ecological concepts. It utilizes appropriate technology and appropriate farming methods. The principles involved in this method are use and development of appropriate technology based up on the understanding of biological system, maintenance of soil fertility for optimum production by using renewable sources.

## Dryland Agriculture

Indian agriculture is predominantly a rainfed agriculture under which both dryfarming and dryland agriculture are included. Out of the 143 million ha of total cultivated area in the country, 101 million ha (i.e. nearly 70%) area are rainfed. In dryland areas, variation in amount and distribution of rainfall influence the crop production as well as socio-economic conditions of farmers. The dryland areas of the country contribute about 42% of the total food grain production. Most of the coarse

grains like sorghum, pearl millet, finger millet and other millets are grown in drylands only. The attention has been paid in the country towards the development of dryland farming. Efforts were made to improve crop yields in research projects at Manjari, Solapur, Bijapur, Raichur and Rohtak. An all India co-ordinated research project for Dryland Agriculture was launched by ICAR in 1970 in collaboration with Government of Canada and later Central Research Institute for Dryland Agriculture (CRIDA) was established at Hyderabad.

**Characteristics of Dryland Agriculture**

Dry land areas may be characterized by the following features,

1. Uncertain, ill-distributed and limited annual rainfall
2. Occurrence of extensive climatic hazards like drought, flood etc.
3. Undulating soil surface
4. Occurrence of extensive and large holdings
5. Practice of extensive agriculture, i.e., prevalence of monocropping etc.
6. Relatively large size of fields
7. Similarity in types of crops raised by almost all the farmers of a particular region
8. Very low crop yield
9. Poor economy of the farmers

**Improved dryland technologies**

Following are the various improved techniques and practices recommended for achieving the objective of increased and stable crop production in dryland areas.

- Crop planning: Crop varieties for dryland areas should be of short duration through resistant tolerant and high yielding which can be harvested within rainfall periods and have sufficient residual moisture in soil profile for post-monsoon cropping.
- Planning for weather: Variation in yields and output of the dryland agriculture is due to the observation in weather conditions especially rainfall. An aberrant weather can be categorized in three types viz.,
    a. Delayed onset of monsoon.
    b. Long gaps or breaks in rainfall and
    c. Early cessation of rains towards the end of monsoon season.
- Crop substitution: Traditional crops/varieties which are inefficient utilizer of soil moisture, less responsive to production input and potentially low producers should be substituted by more efficient ones.

- Cropping systems: Increasing the cropping intensities by using the practice of intercropping and multiple cropping is the way of more efficient utilization of resources. The cropping intensity would depend on the length of growing season, which in turn depends on rainfall pattern and the soil moisture storage capacity of the soil.
- Fertilizer use: The availability of nutrients is limited in drylands due to the limiting soil moisture. Therefore, application of the fertilizers should be done in furrows below the seed. The use of fertilizers is not only helpful in providing nutrients to crop but also, helpful in efficient use of soil moisture. A proper mixture of organic and inorganic fertilizers improves moisture holding capacity of soil and increase during tolerance.
- Rain water management: Efficient rain water management can increase agricultural production from dryland areas. Application of compost and farm yard manure and raising legumes add the organic matter to the soil and increase the water holding capacity. The water, which is not retained by the soil, flows out as surface runoff. This excess runoff water can be harvested in storing dugout ponds and recycled to donar areas in the server stress during rainy season or for raising crops during winter.
- Watershed management: Watershed management is a approach to optimize the use of land, water and vegetation in a area and thus, to provide solution drought, moderate floods, prevent soil erosion, improve water availability and increase fuel, fodder and agricultural production on a sustained basis.
- Alternate Land use: All drylands are not suitable for crop production. Same lands may be suitable for range/ pasture management and for tree farming and ley farming, dryland horticulture, agro-forestry systems including alley cropping. All these systems which are alternative to crop production are called as alternate land use systems. This system helps to generate off-season employment mono-cropped dryland and also, minimizes risk, utilizes off-season rains, prevents degradation of soils and restores balance in the ecosystem. The different alternate land use systems are alley cropping, agri-horticultural systems and silvi-pastoral systems, which utilizes the resources in better way for increased and stabilized production from drylands.







